Data-Oriented Debugging Across Place and Time

Jeffrey Tao, Xiaorui Liu, Ryan Marcus, Andrew Head

UIST 2026

Do you use print debugging?

Do you use print debugging?

Do you use breakpoint debugging?

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Do you use breakpoint debugging?

Why?

Data-Oriented Debugging

This is an unconventional talk!

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1. Conceptual Lens: Data-Oriented Debugging
2. Observing Data-Oriented Debugging in studies
3. Future Data-Oriented Directions

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1. Conceptual Lens: Data-Oriented Debugging
2. Observing Data-Oriented Debugging in studies
3. Future Data-Oriented Directions

There will be no AI! But ask during Q&A 😊

Calculator Example

```
def evaluate(node):
    if isinstance(node, Num):
        return node.value
    left = evaluate(node.left)
    right = evaluate(node.right)
    match node.op:
        case "+": result = left + right
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Terminal

```
>>> calculate("2 + 3 * 4 ^ 2")
26
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"Order of operations is incorrect?"

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"Nope!"

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"Nope!"

"Order of operations is incorrect?"

① Data Need

Calculator Example

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6     match node.op:
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9         case "^": result = left * right
10     print(node.op)          ② Query
11     return result
```

Terminal

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② Query

Terminal

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③ **Interpret output**

"Nope!"

"Order of operations is incorrect?"

① Data Need

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② Query

Terminal

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```

③ Interpret output

"Nope!"

④ Update mental model

"Order of operations is incorrect?"

① Data Need

Calculator Example

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② Query

Takeaway:
Think about data needs and how queries are specified

Terminal

"Order of operations is incorrect?"

① Data Need

Place & Time

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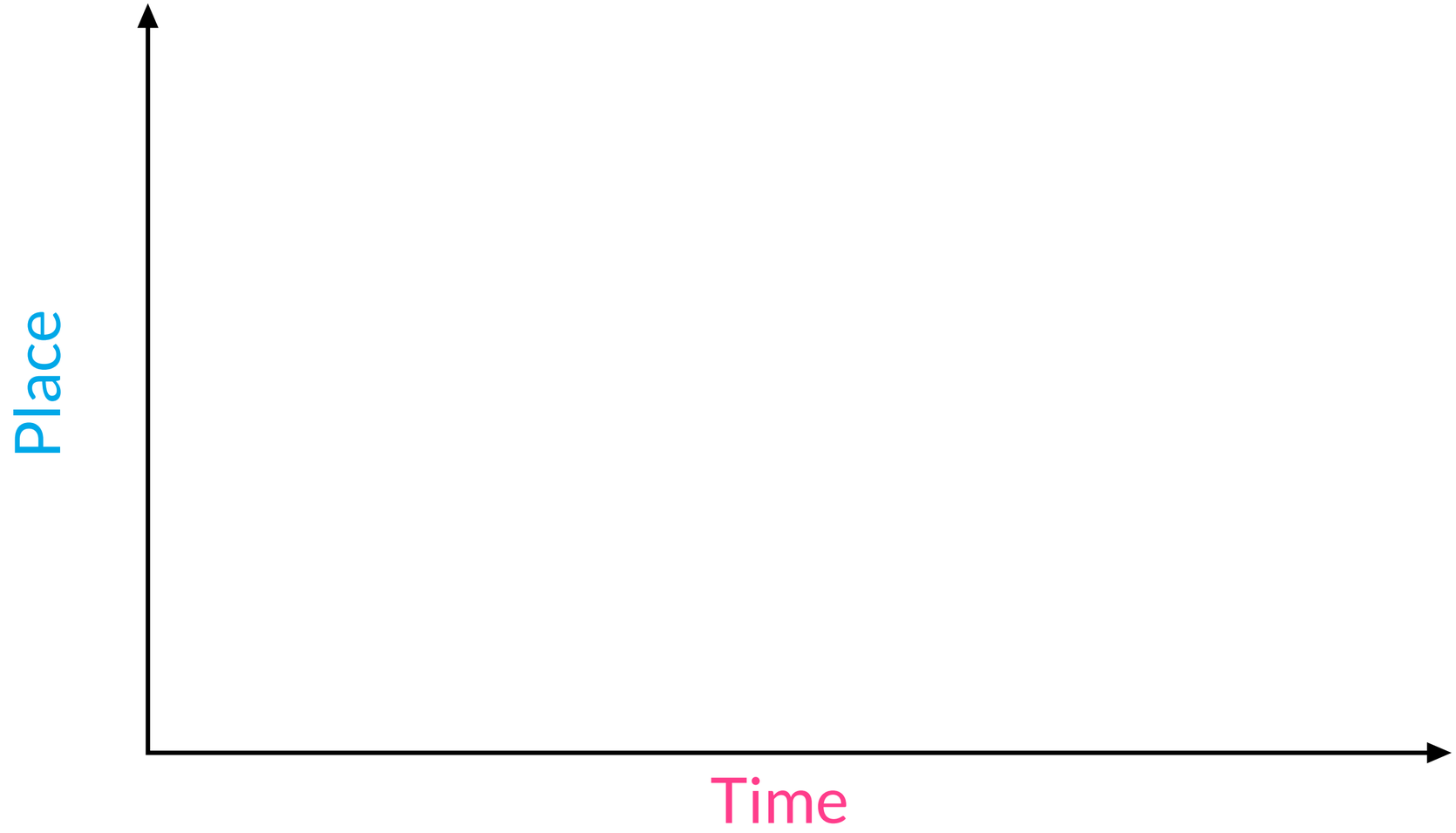
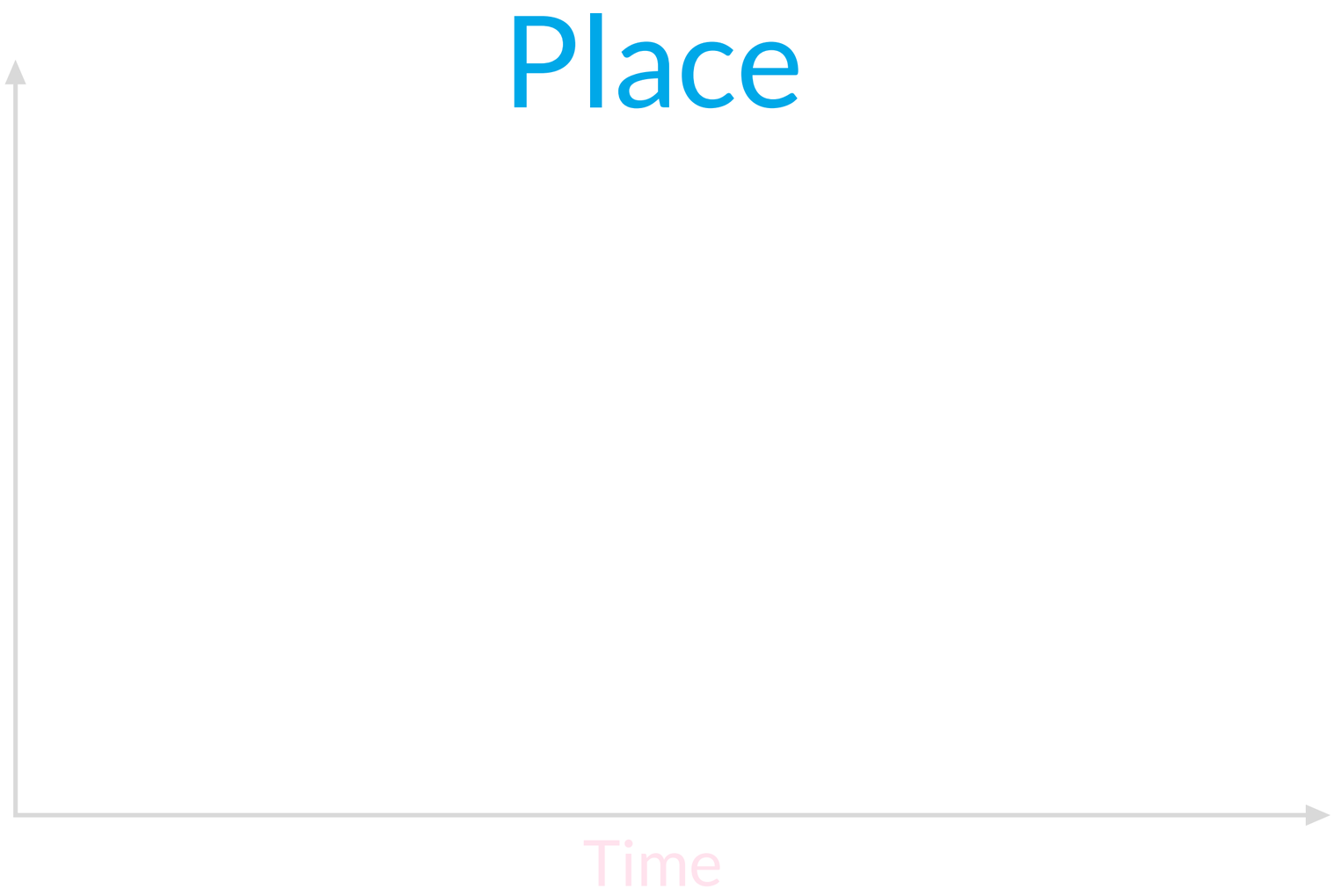
Place & Time



Place & Time



Place & Time

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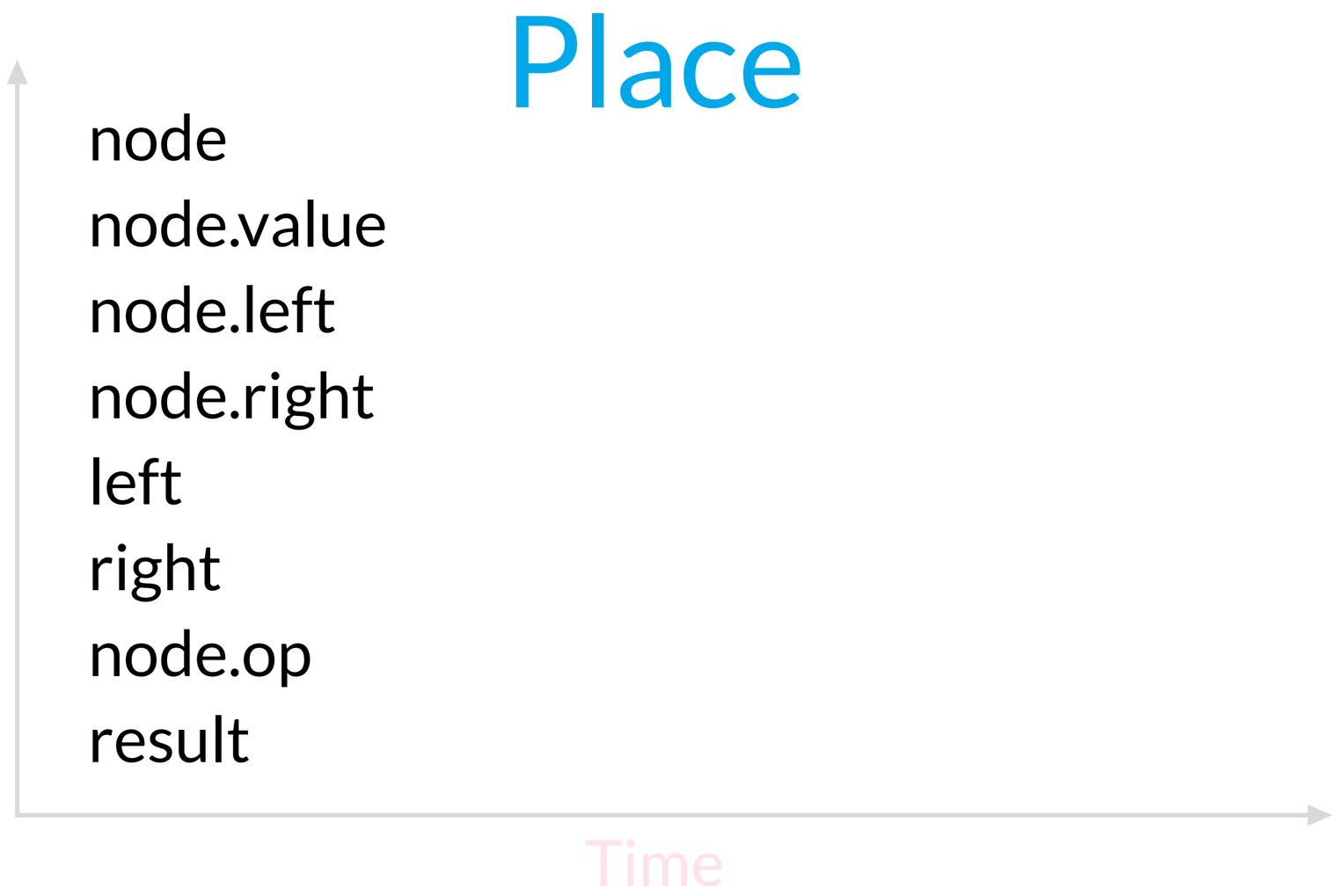
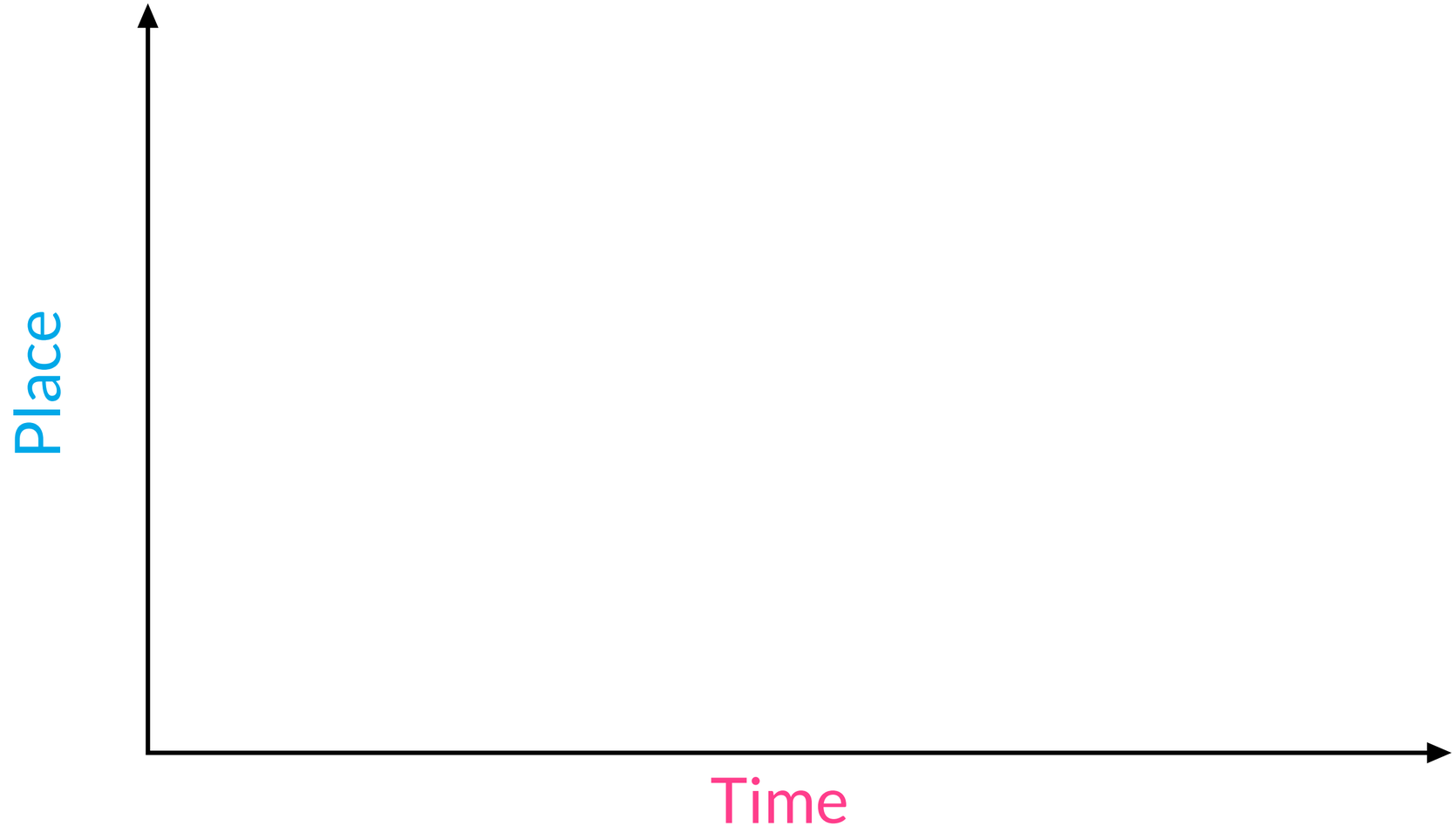
Time

Place & Time


Place
node
node.value
node.left
node.right
left
right
node.op
result
Time

Place & Time



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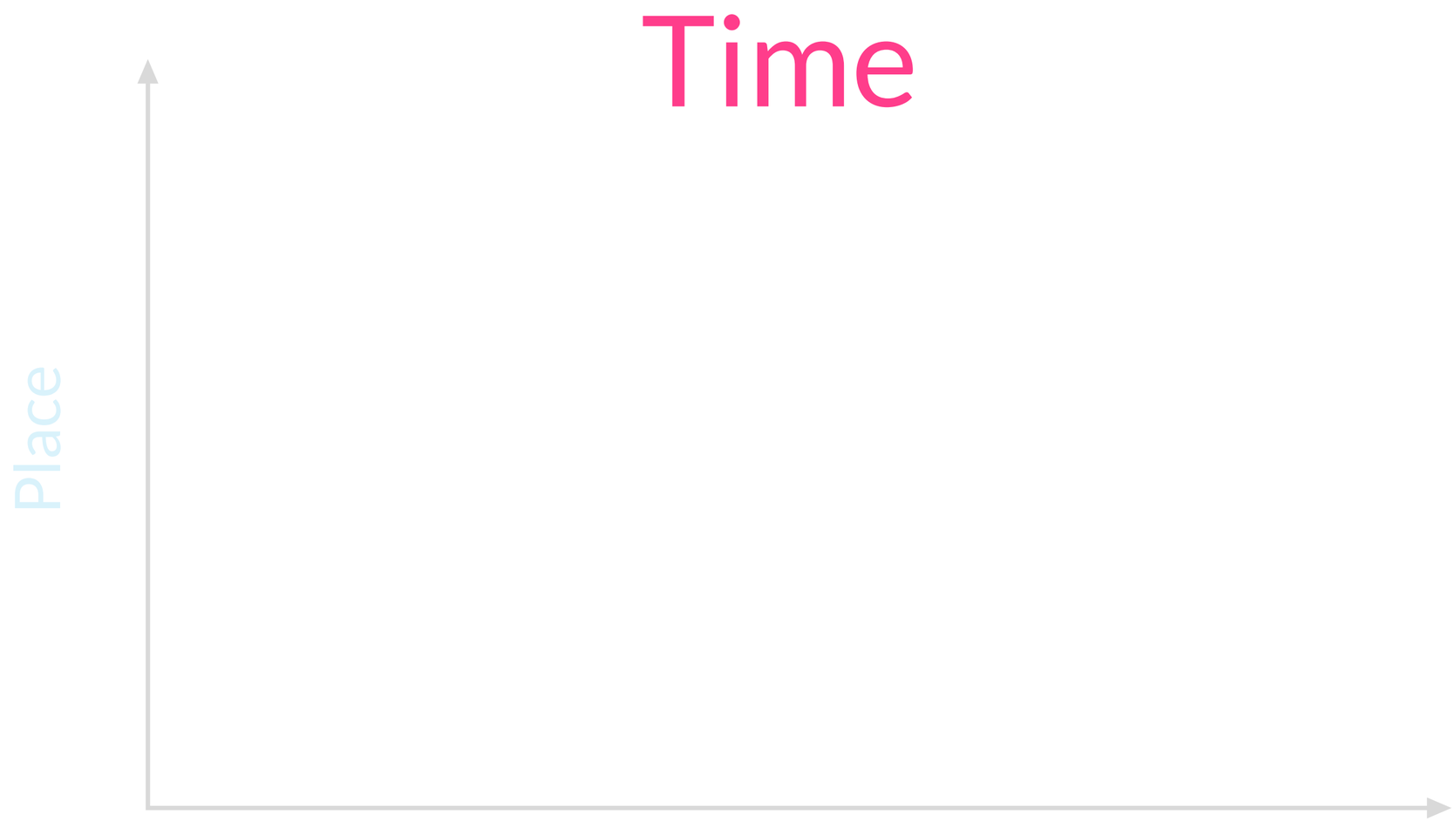

Place & Time

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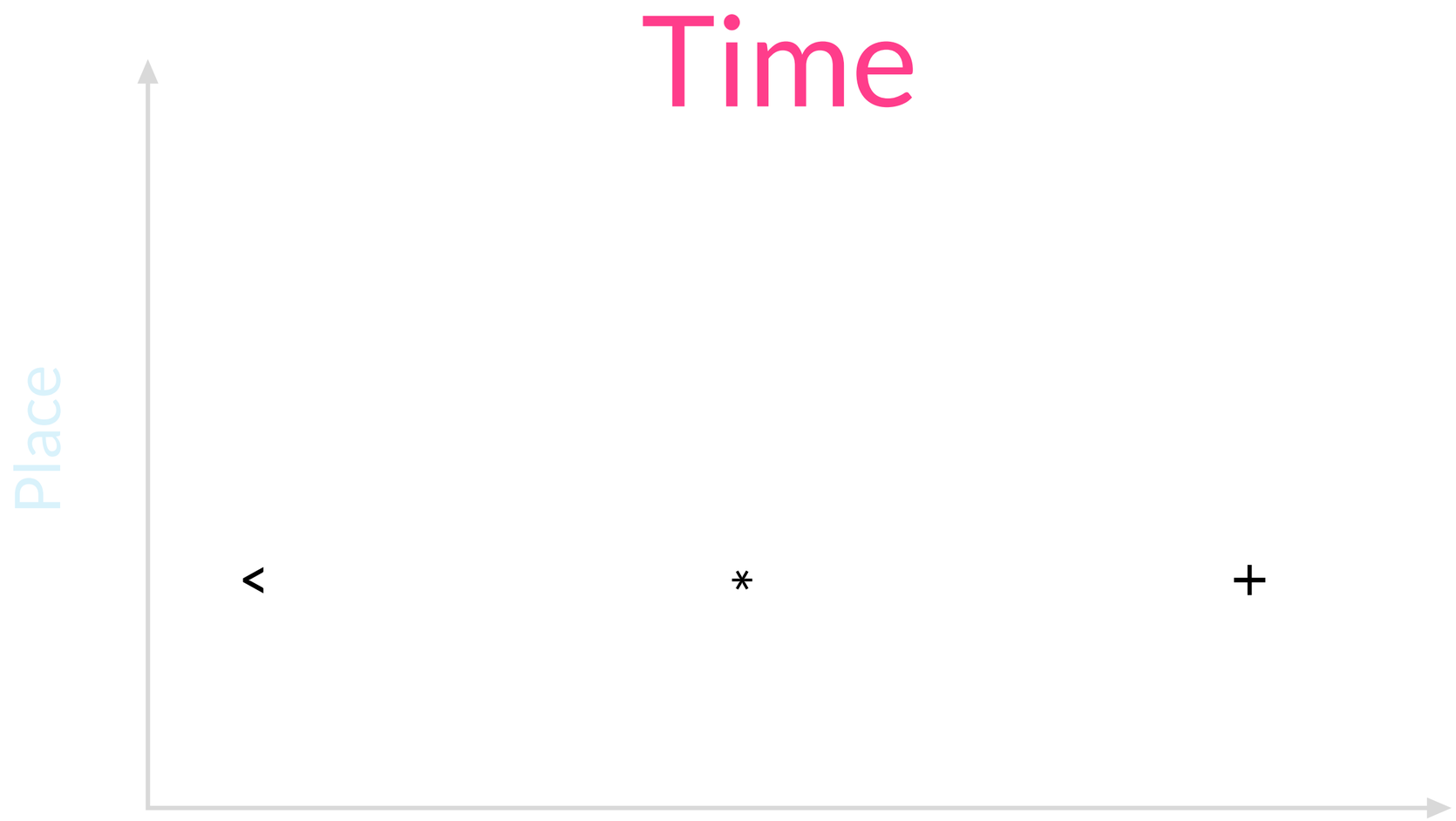
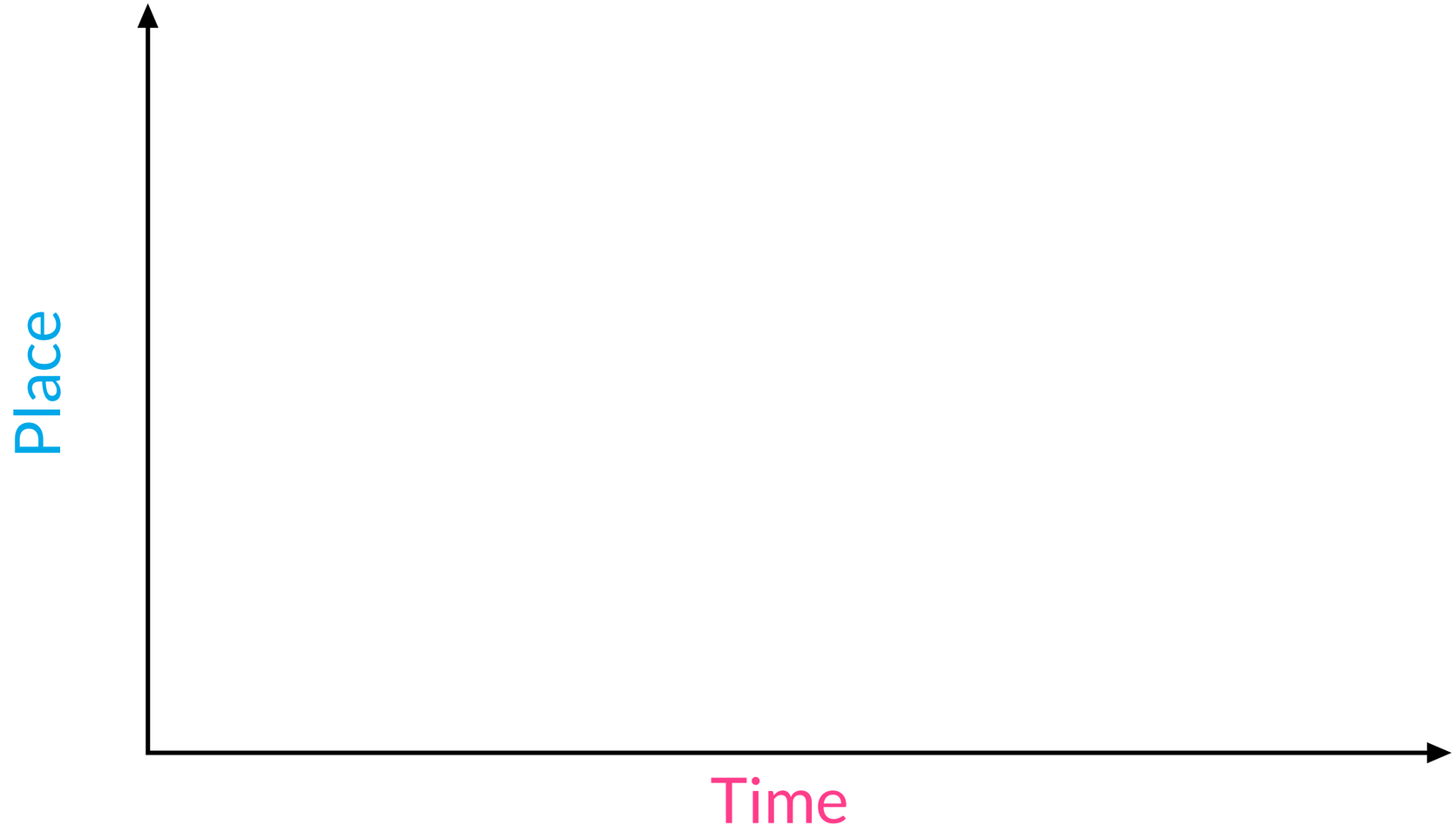


Place & Time



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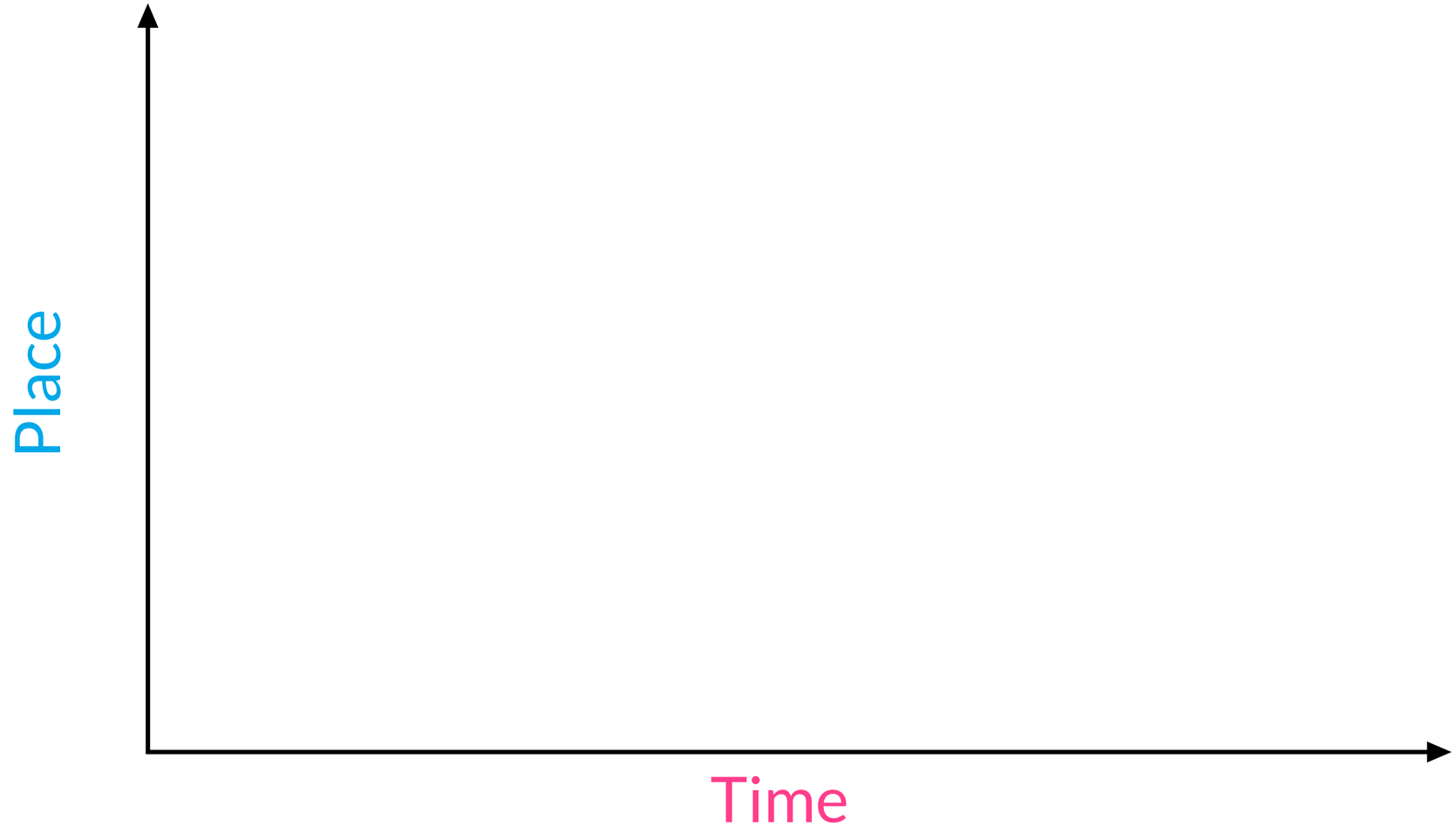
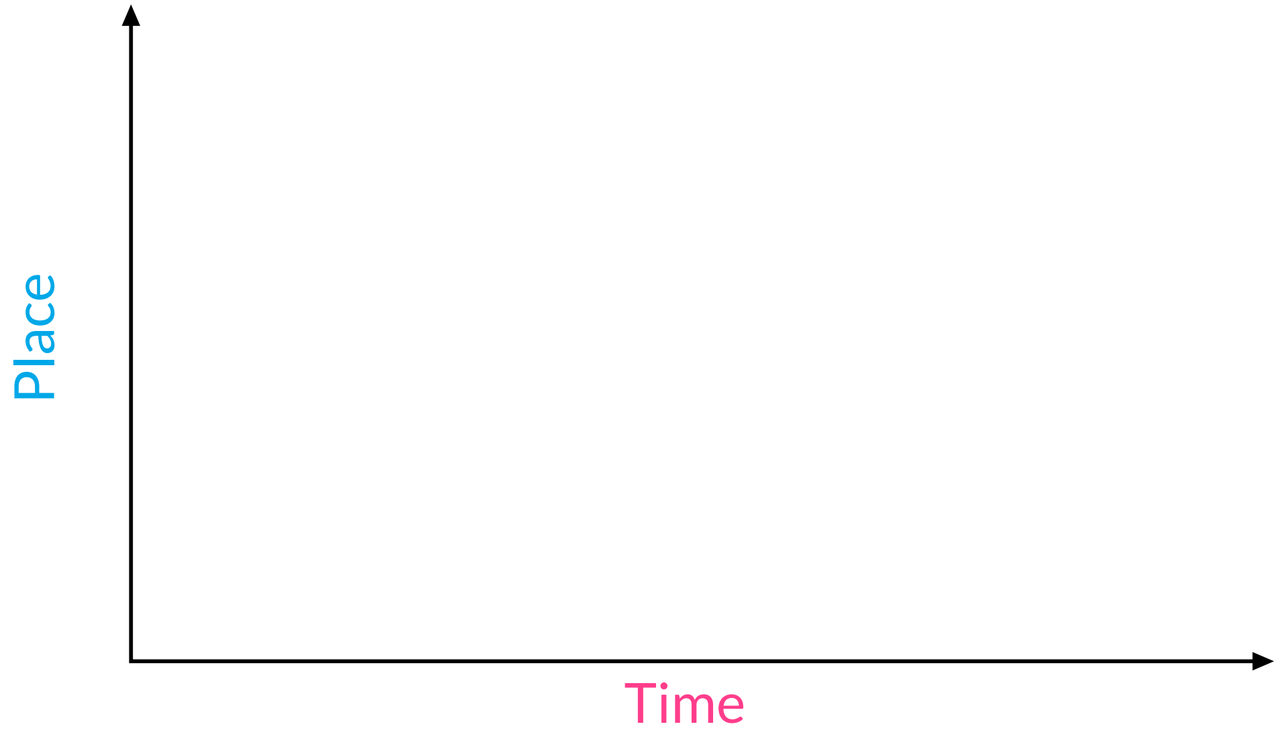


Place & Time

print debugging

Some Place · Across Time



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Place & Time

print debugging

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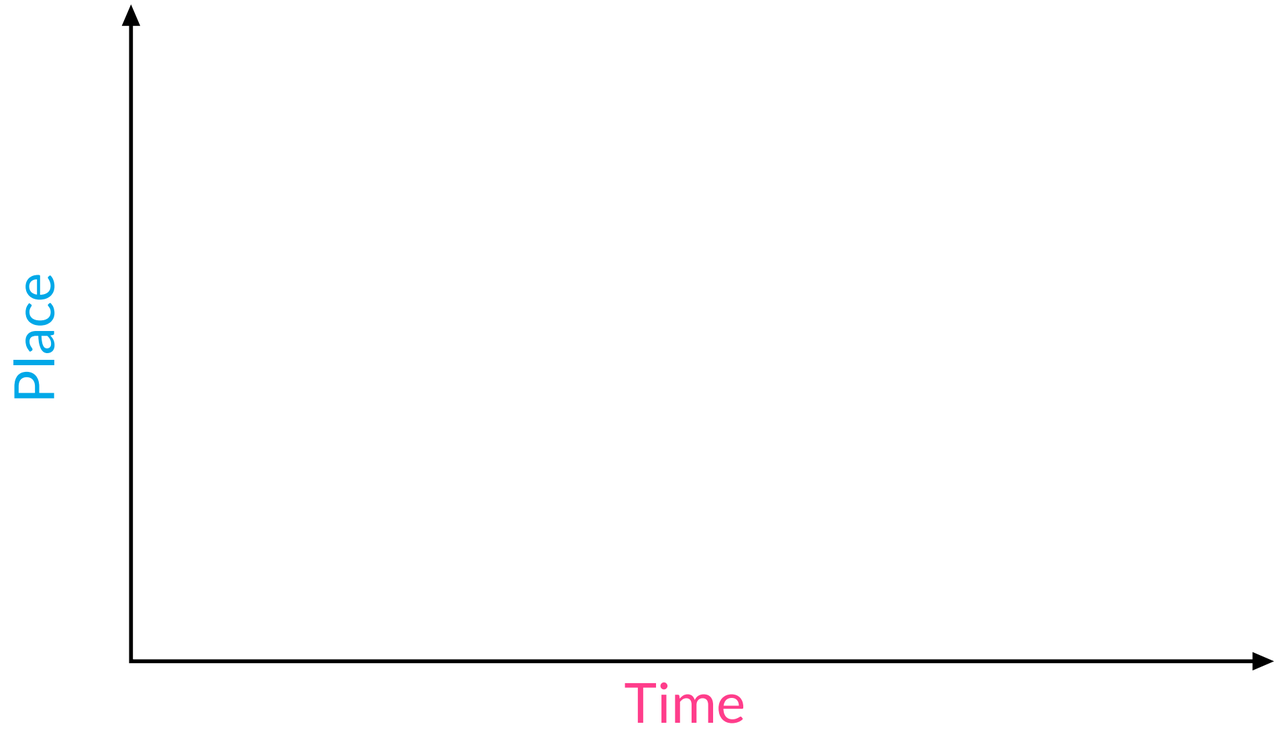

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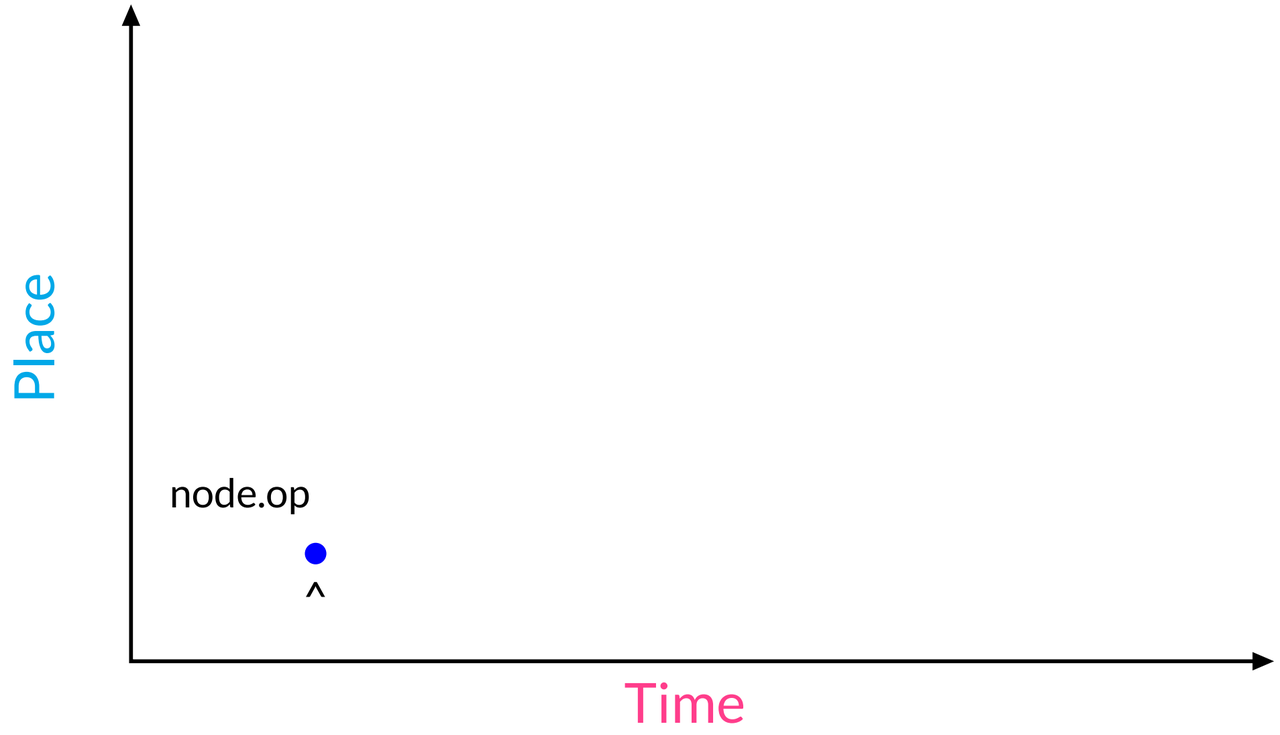


```
evaluate(node) · 4 ^ 2 → 8
1 def evaluate(node):
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```
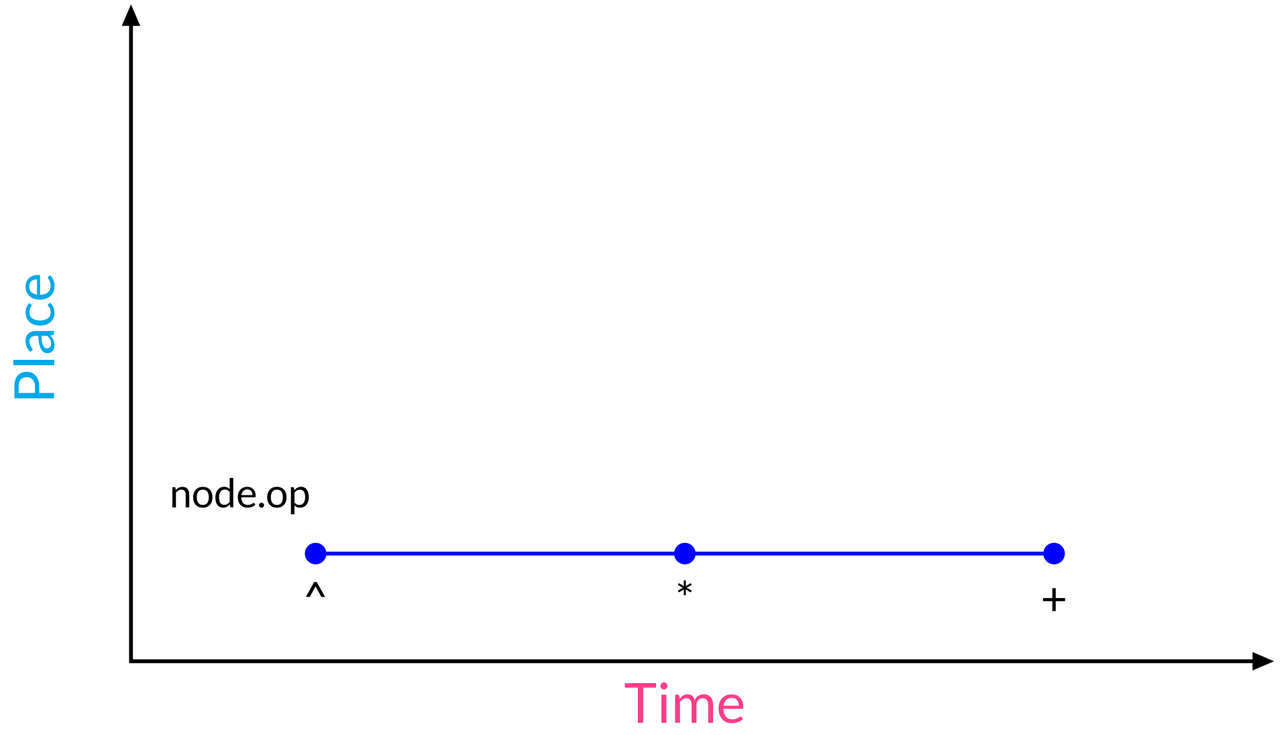

Place & Time

print debugging

Some Place · Across Time



evaluate(node) · 2 + 24 → 26

```python
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     print(node.op)
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```

Terminal

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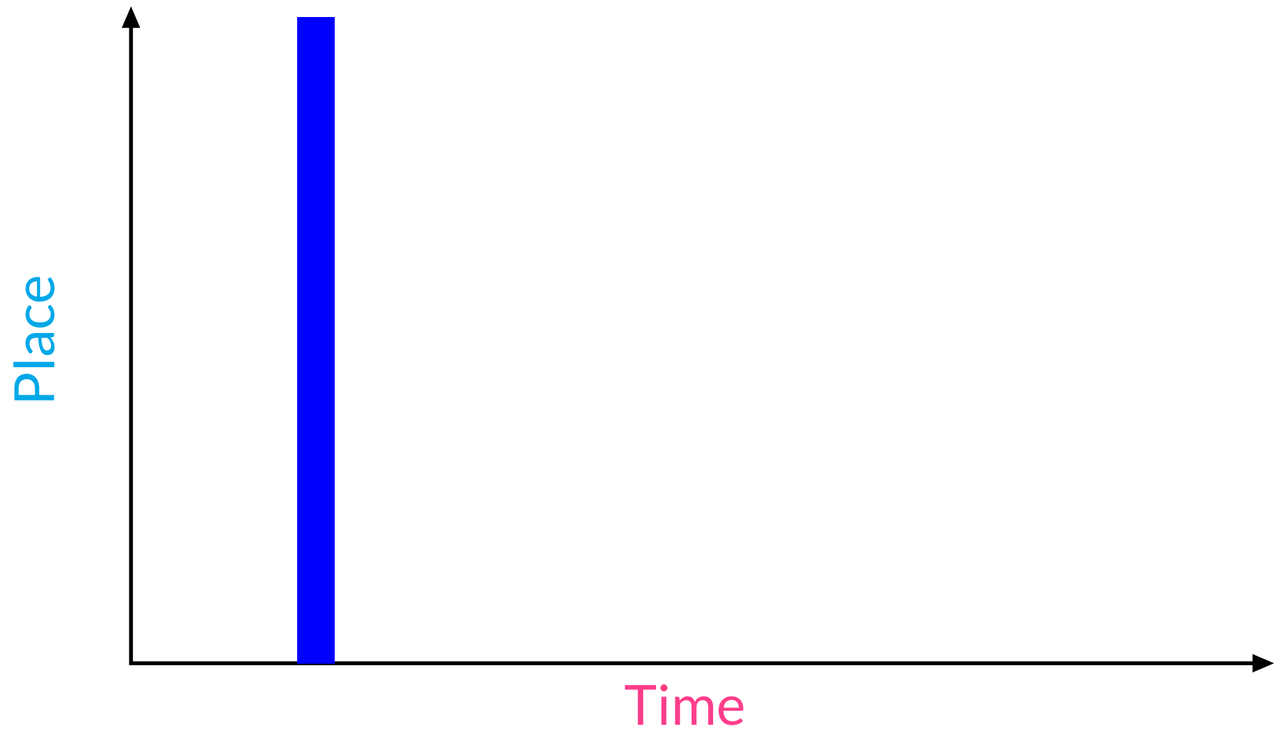
```

Place & Time

breakpoint debugging

Across Place · Point in Time

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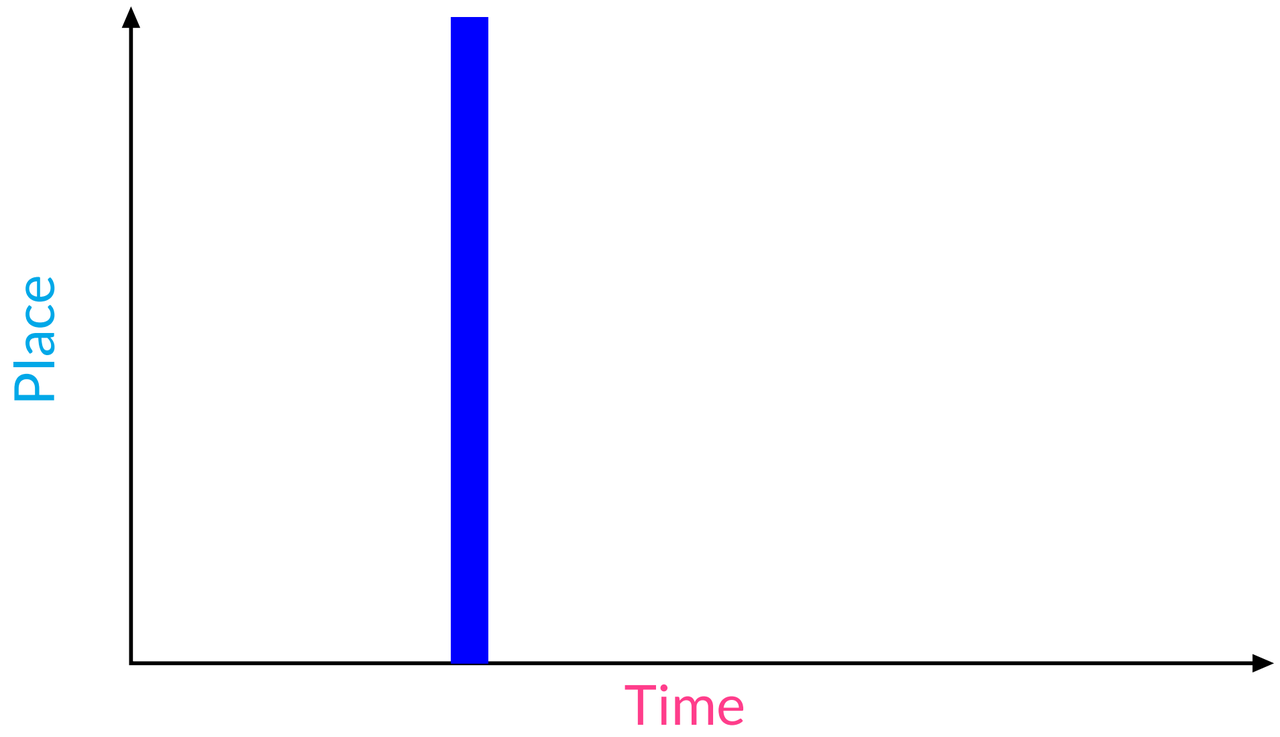
Variables · evaluate

node	BinOp("^")	node.value	–
node.left	Num(4)	node.right	Num(2)
left	4	right	2
node.op	"^"	result	not assigned

Place & Time

breakpoint debugging

Across Place · Point in Time

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Variables · evaluate

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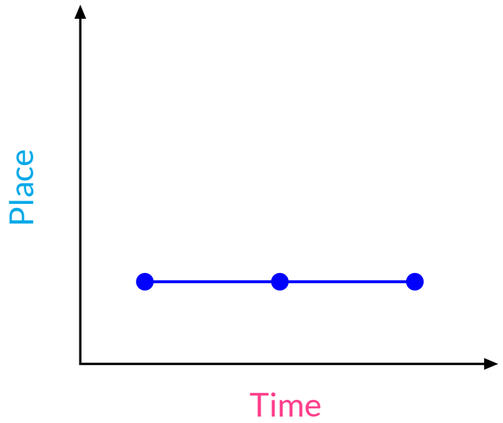
node.left	Num(4)	node.right	Num(2)
left	4	right	2
node.op	"^"	result	8

Debuggers Under The Lens



Print debugging

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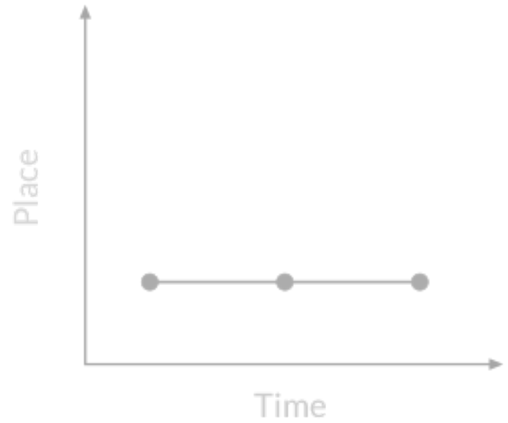

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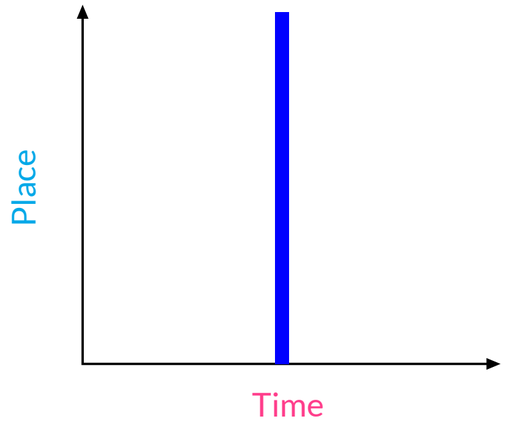
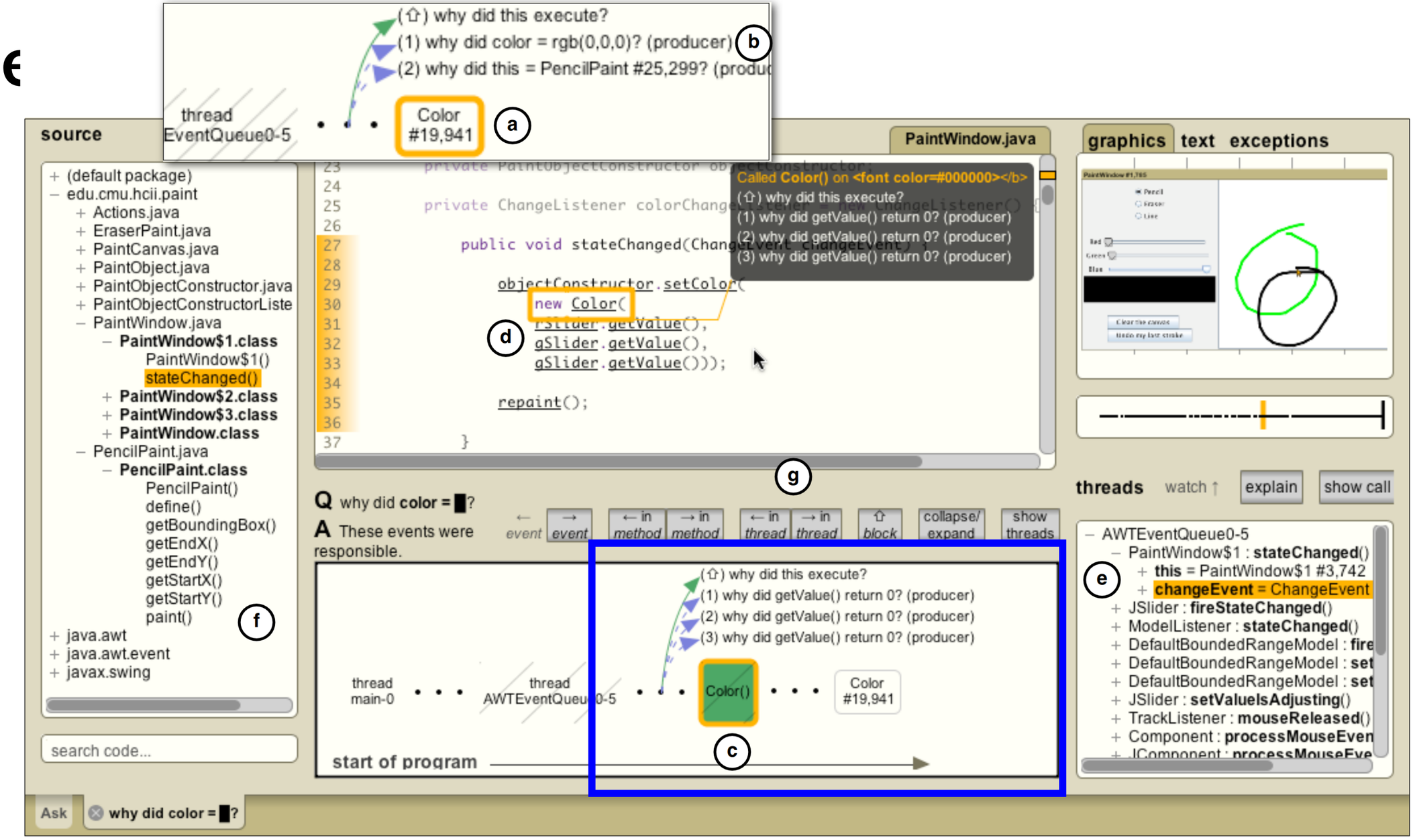
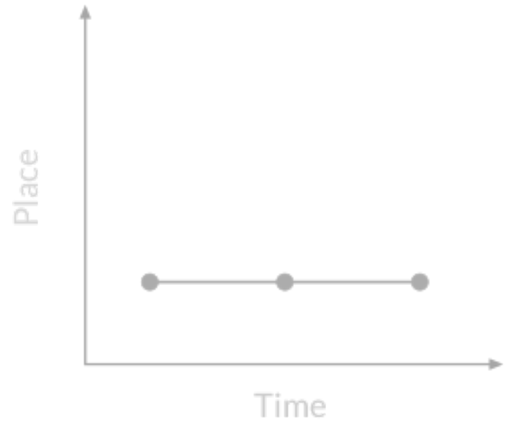


Breakpoint debugging

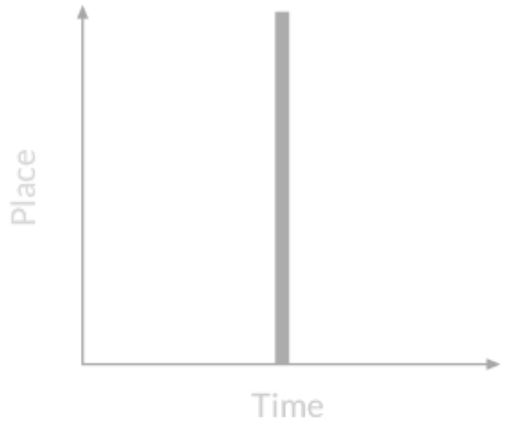
De


(⇧) why did this execute?
(1) why did color = rgb(0,0,0)? (producer)
(2) why did this = PencilPaint #25,299? (produc
b
thread
EventQueue0-5
Color
#19,941
a
source
PaintWindow.java
graphics
text
exceptions
+ (default package)
– edu.cmu.hcii.paint
+ Actions.java
+ EraserPaint.java
+ PaintCanvas.java
+ PaintObject.java
+ PaintObjectConstructor.java
+ PaintObjectConstructorListe
– PaintWindow.java
– PaintWindow$1.class
PaintWindow$1()
stateChanged()
+ PaintWindow$2.class
+ PaintWindow$3.class
+ PaintWindow.class
– PencilPaint.java
– PencilPaint.class
PencilPaint()
define()
getBoundingBox()
getEndX()
getEndY()
getStartX()
getStartY()
paint()
+ java.awt
+ java.awt.event
+ javax.swing
f
search code...
Called Color() on <font color=#000000>
(⇧) why did this execute?
(1) why did getValue() return 0? (producer)
(2) why did getValue() return 0? (producer)
(3) why did getValue() return 0? (producer)
objectConstructor.setColor(
new Color(
gSlider.getValue(),
gSlider.getValue(),
gSlider.getValue()));
repaint();
d
g
Q why did color = ?
A These events were responsible.
← event
→ event
← in method
→ in method
← in thread
→ in thread
⇧ block
collapse/ expand
show threads
thread
main-0
thread
AWTEventQueue-0-5
Color()
Color
#19,941
c
start of program
threads
watch ↑
explain
show call
– AWTEventQueue0-5
– PaintWindow$1 : stateChanged()
+ this = PaintWindow$1 #3,742
+ changeEvent = ChangeEvent
+ JSlider : fireStateChanged()
+ ModelListener : stateChanged()
+ DefaultBoundedRangeModel : fire
+ DefaultBoundedRangeModel : set
+ DefaultBoundedRangeModel : set
+ JSlider : setValueIsAdjusting()
+ TrackListener : mouseReleased()
+ Component : processMouseEven
e
Ask
why did color = ?

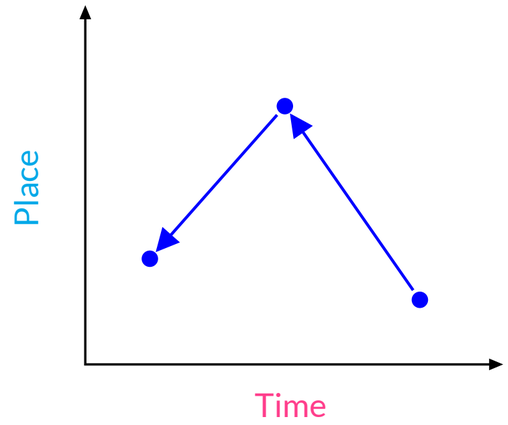
Debuggers Under The Lens



Print debugging

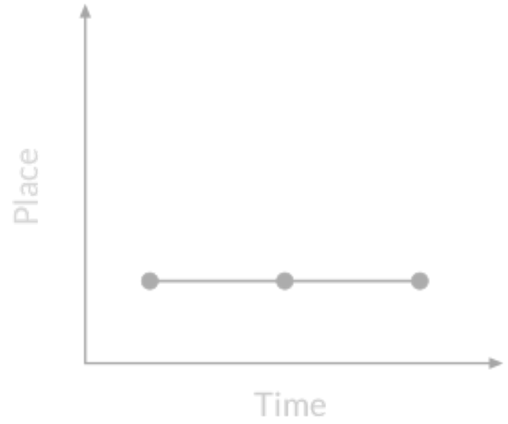


Breakpoint debugging



Whyline

Debuggers Under The Lens



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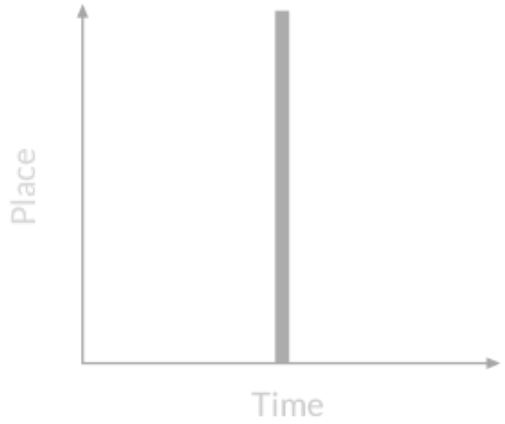
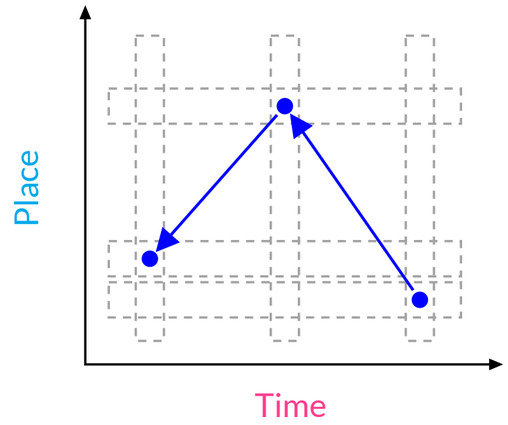


Breakpoint debugging



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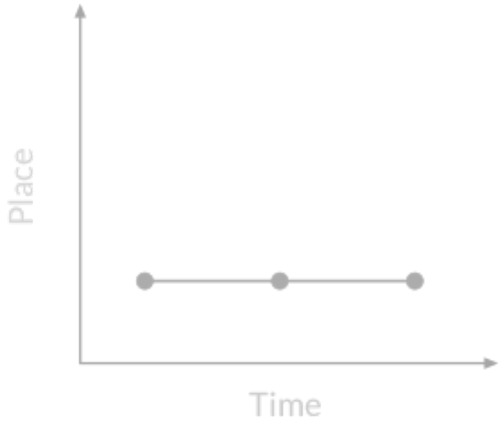

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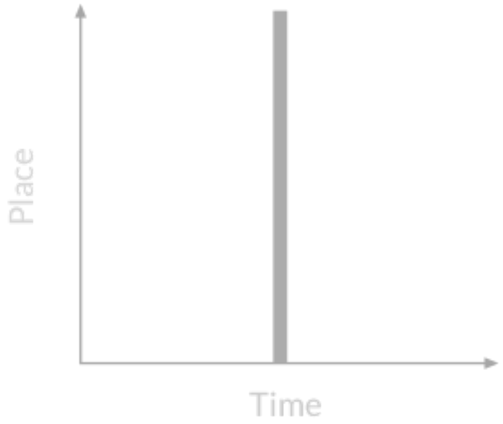

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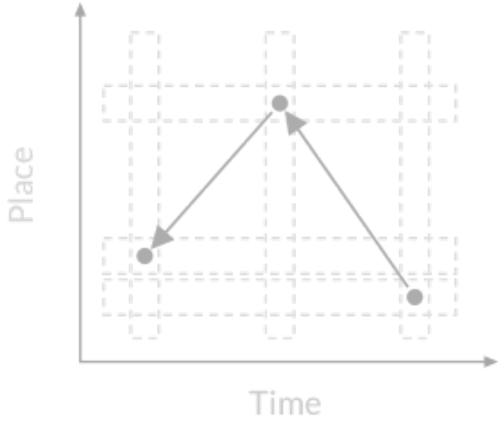
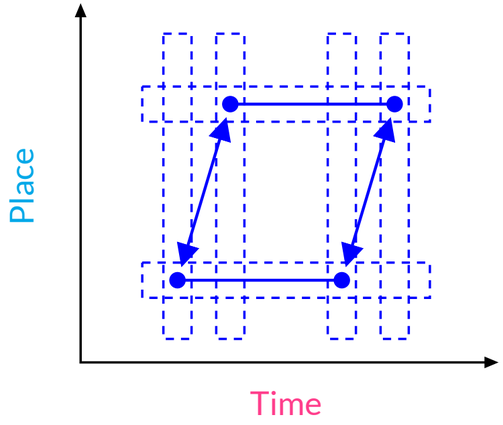


Whyline



Future Tools

Study Findings

Formative Interviews, Comparative Tool-Use Lab Study

Formative Interviews

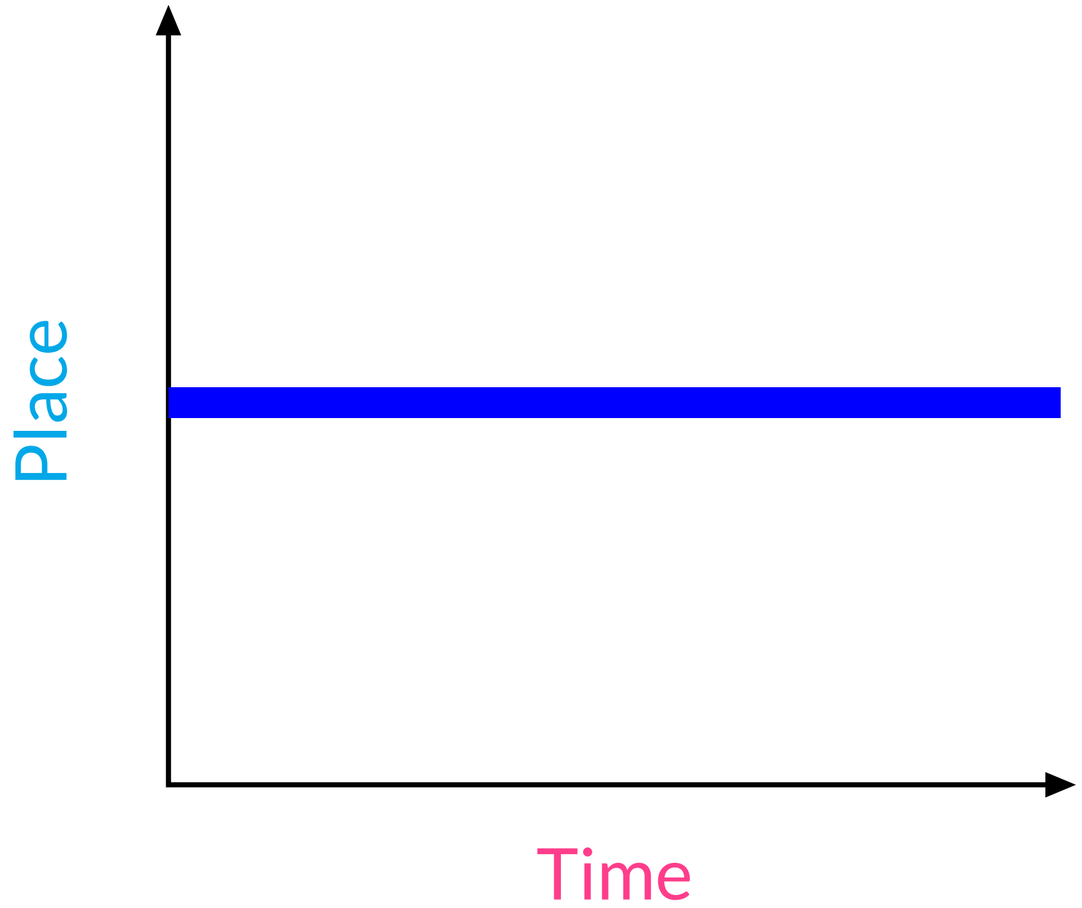
12 industry software engineers, *"How do you debug the systems you work on?"*

Comparative Tool-Use Lab Study

19 undergrad + grad students, 2 debugging tasks, 3 debugging tools

Types of Queries

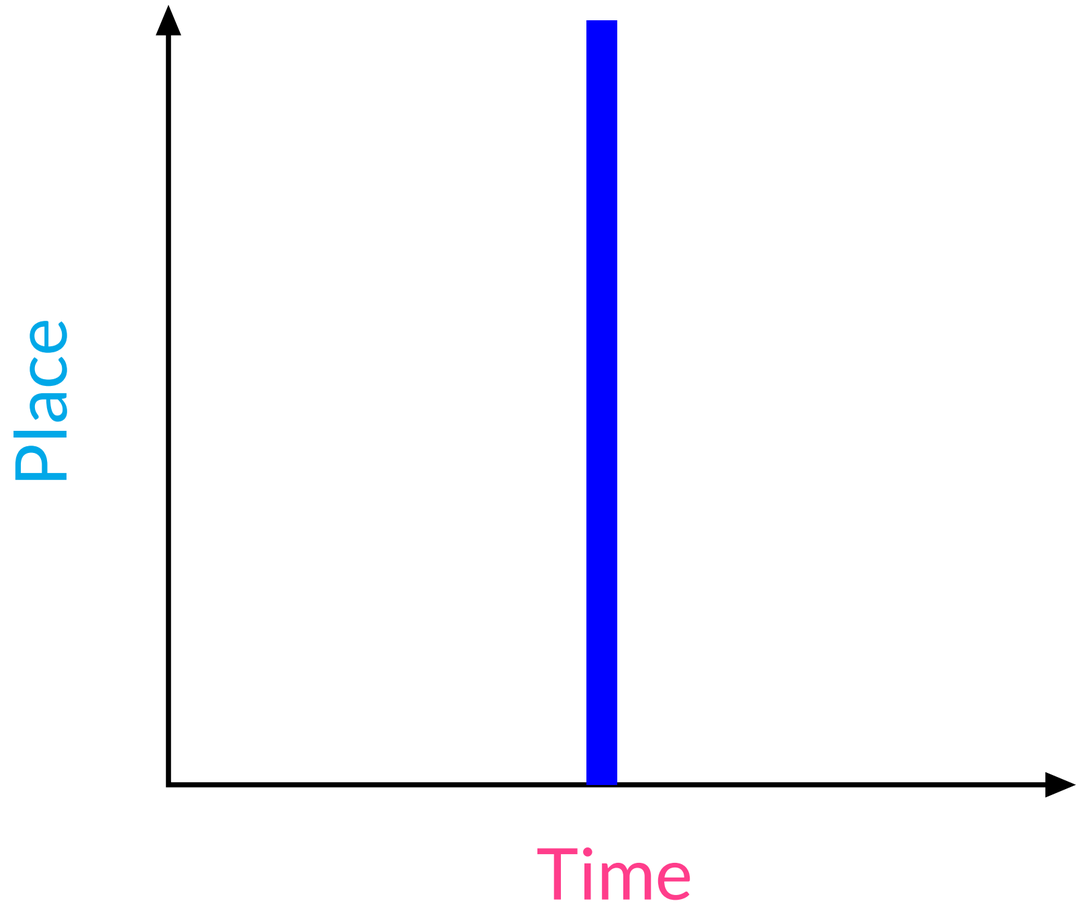
Across Time (17 participants, 58 instances)



"What are all of the values this can have?"

Types of Queries

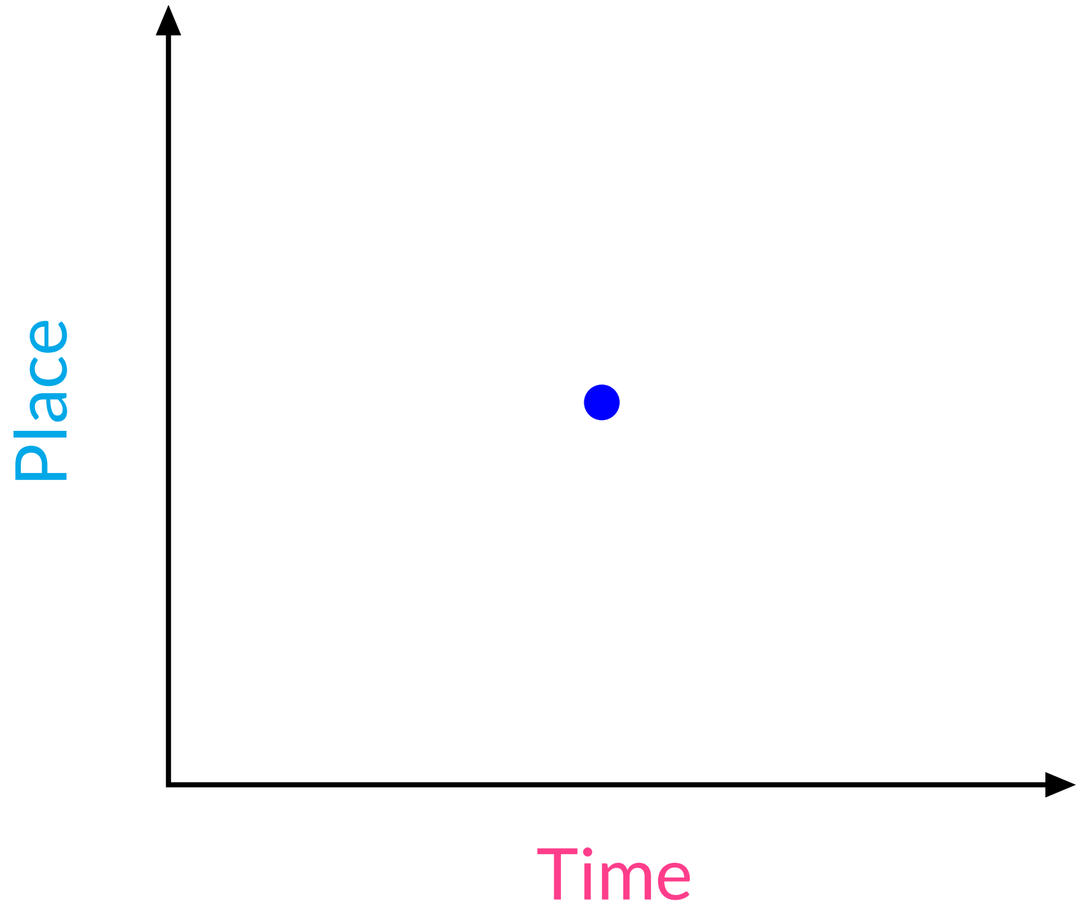
Across Place (6 participants, 9 instances)



"What else is here?"

Types of Queries

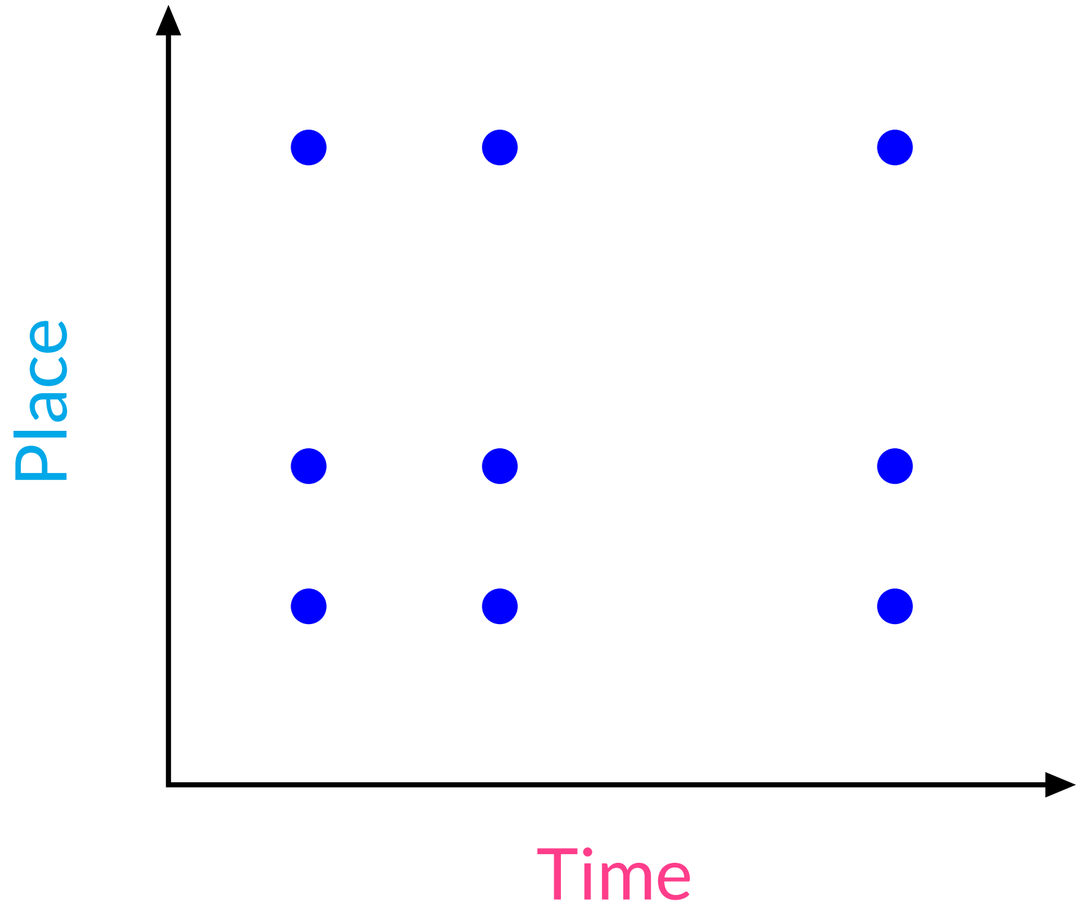
Point in Place and Time (7 participants, 12 instances)



"Did this ever happen?"

Types of Queries

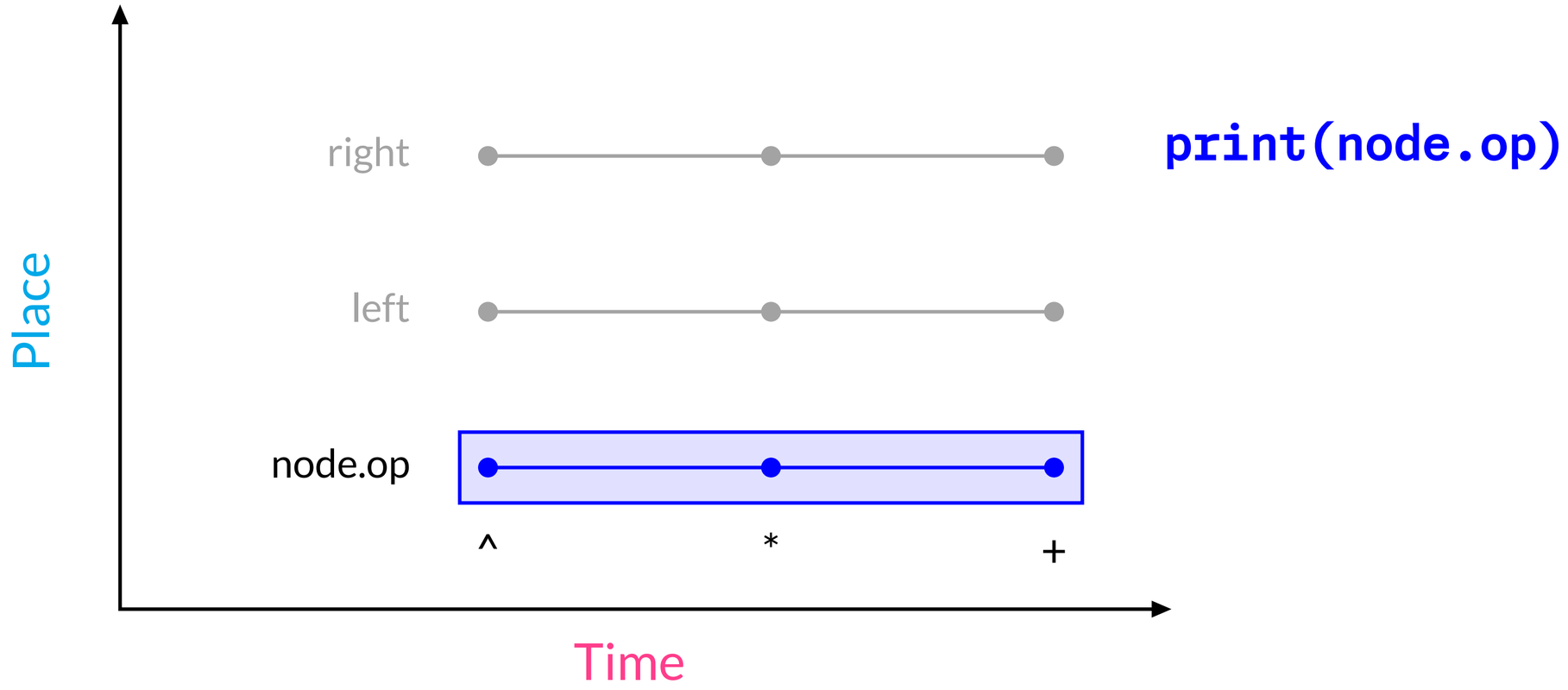
Coordinating Across Place and Time (7 participants)



`log(middleware, response)`

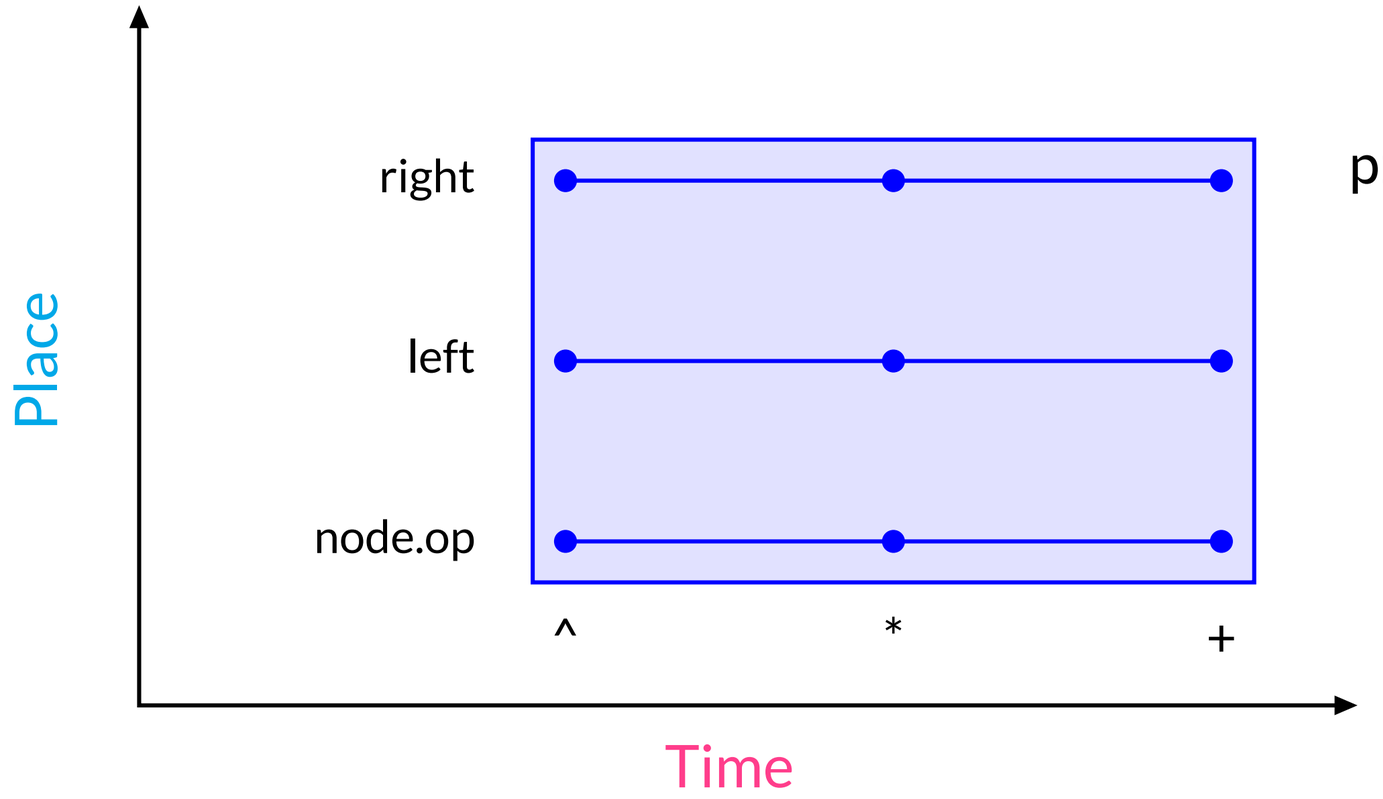
"When the request is rejected, which middleware was running? Oh, a different number of middleware run per-request and multiple different middleware cause rejections."

Transformations



Transformations

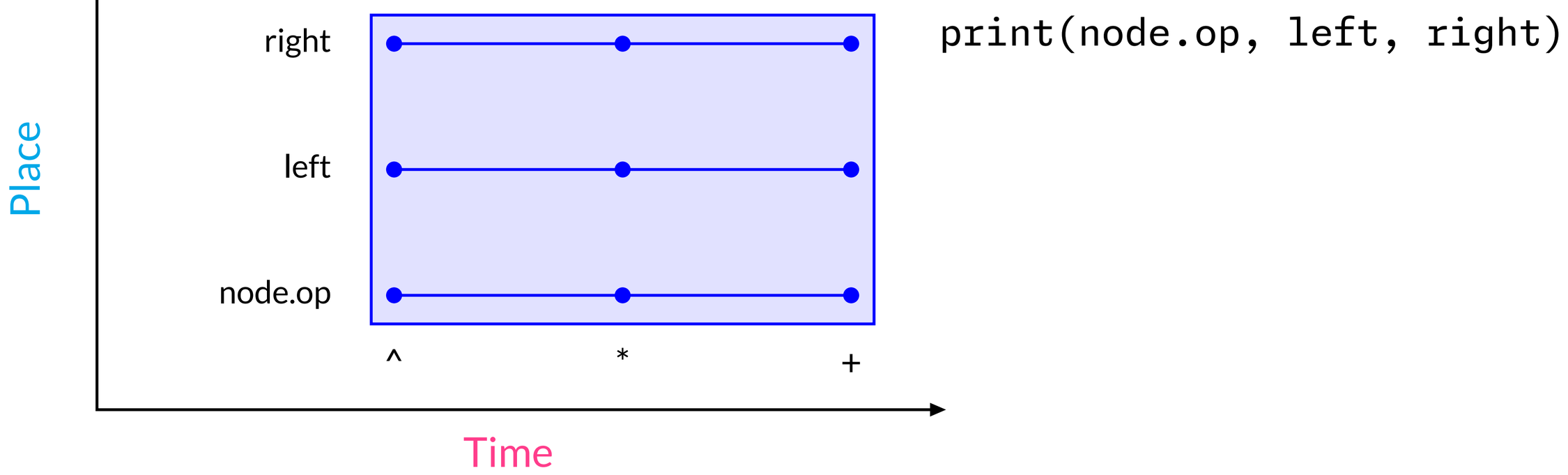
Widen across place (14 participants, 27 instances)




```
print(node.op, left, right)
```

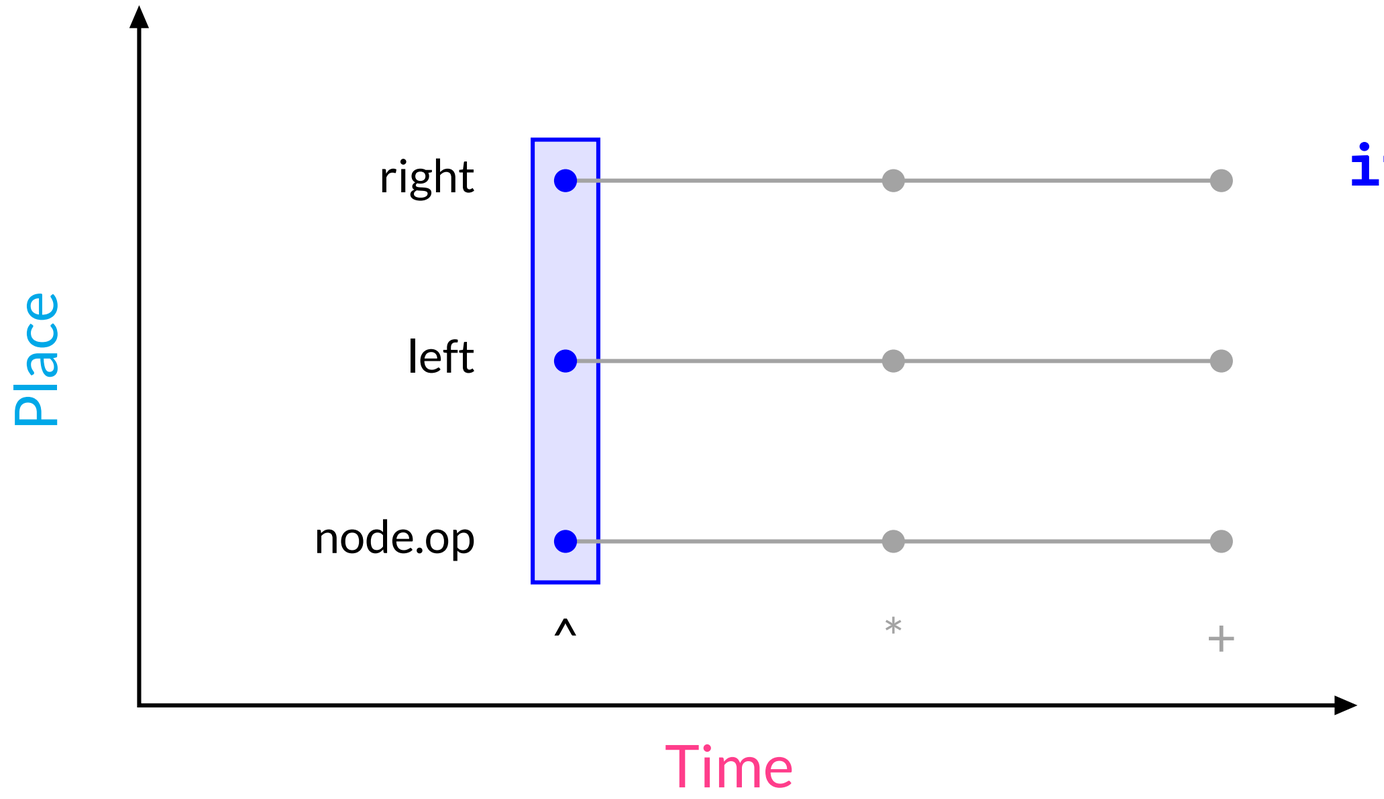
Transformations

Takeaway:
Do you support transformations? Via what interaction?



Transformations

Narrow in time (12 participants, 18 instances)




```
if node.op == "^":
    print(node.op, left, right)
```

Pre-Filtering

Pre-Filtering

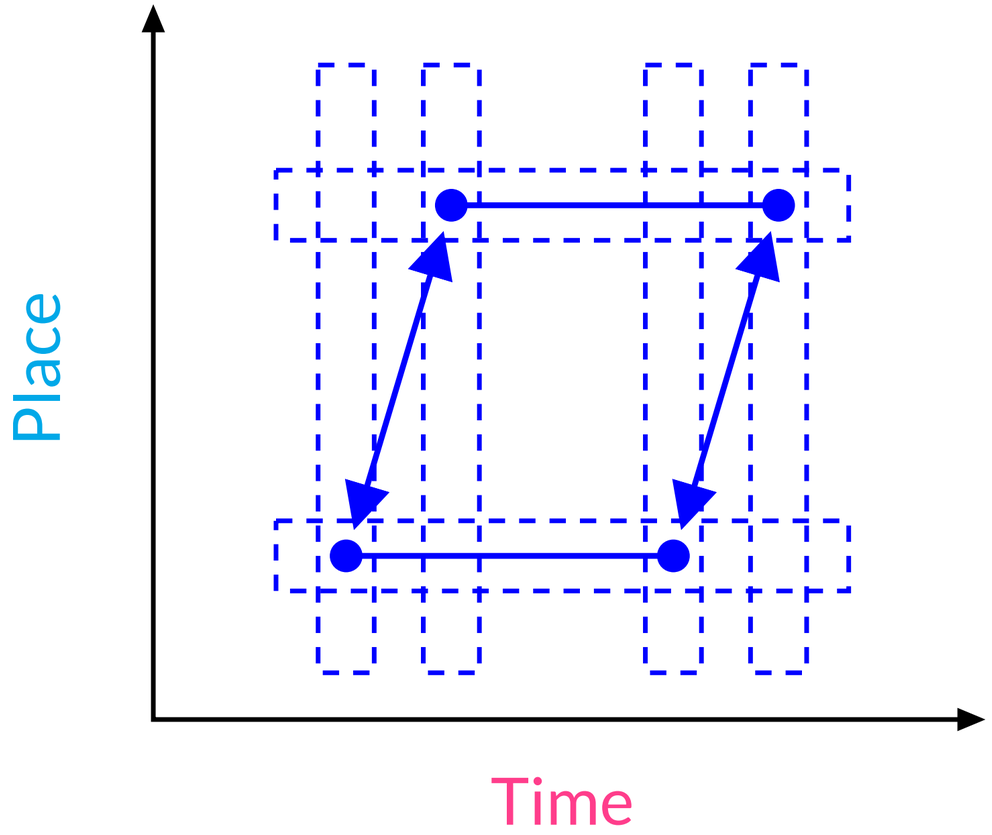
Programmers cope with tool limitations by asking for less

Pre-Filtering

Programmers cope with tool limitations by asking for less

"there will be a lot of statements, right? ...it looks like too much information for me to look at and process and get an output"

The Future


Place
Time

```python
def evaluate(node):
    if isinstance(node, Num):
        return node.value
    left = evaluate(node.left)
    right = evaluate(node.right)
    match node.op:
        case "+": result = left + right
        case "*": result = left * right
        case "^": result = left * right
     return result
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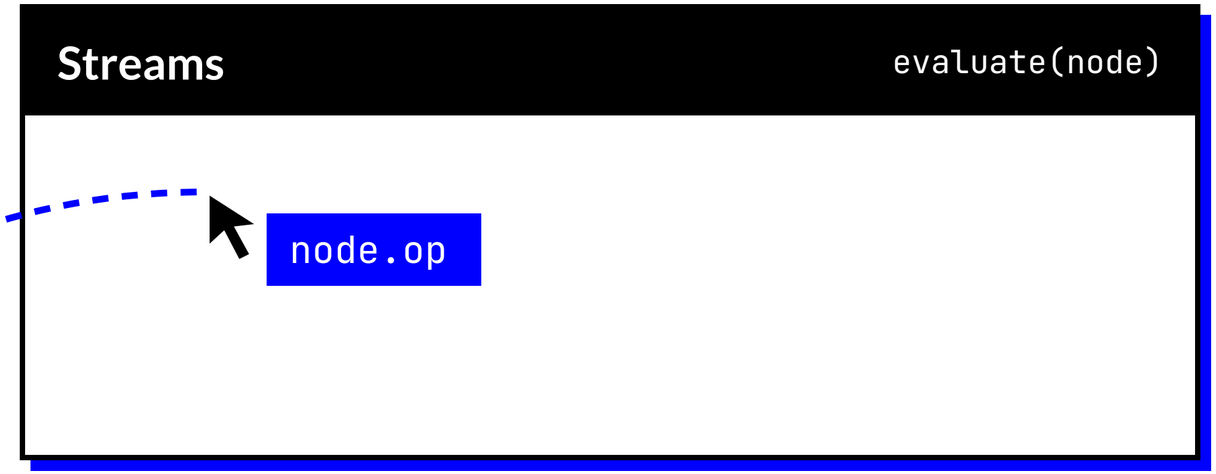

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     return result
```

Streams `evaluate(node)`

node.op
^
*
+

```python
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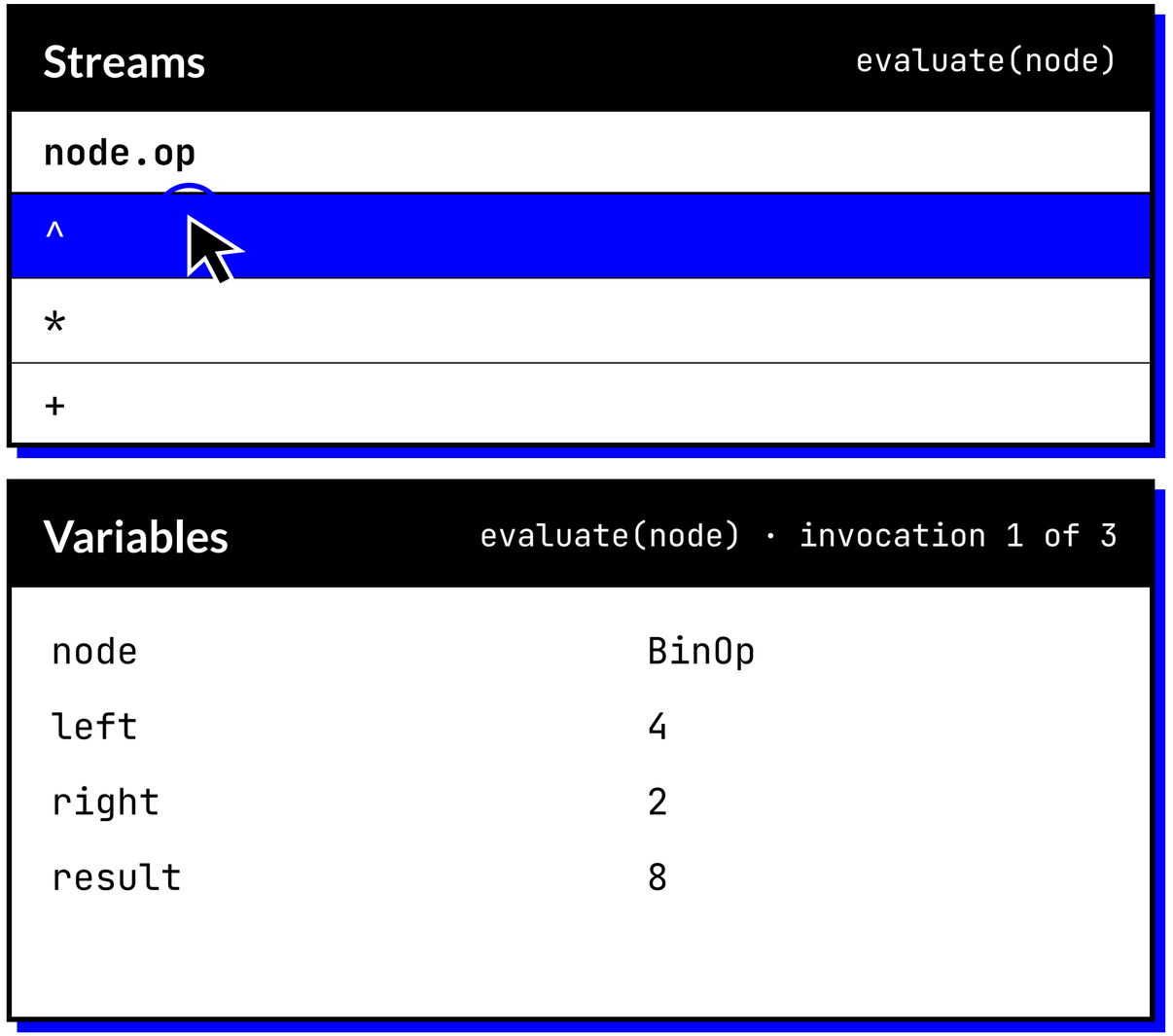

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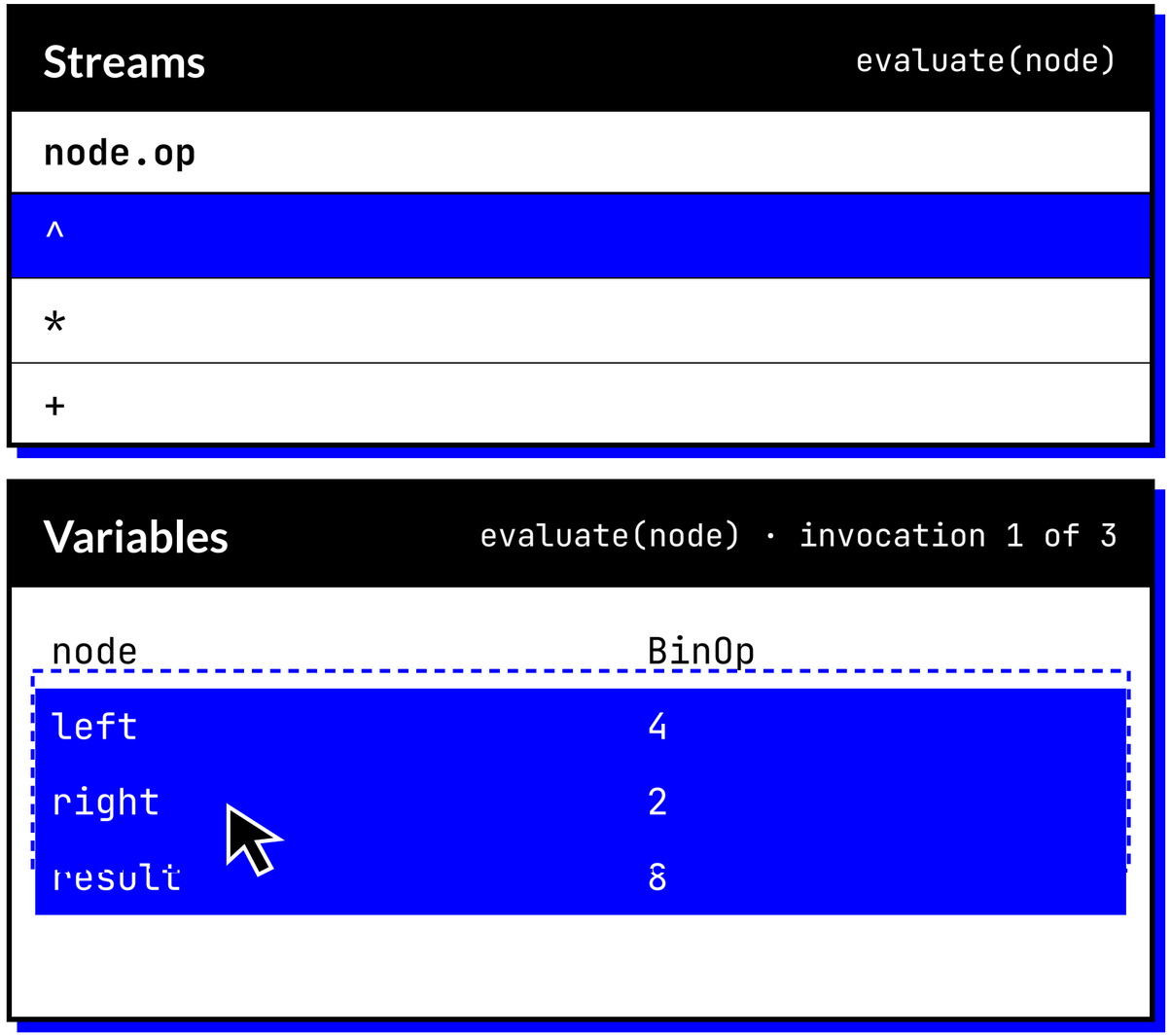

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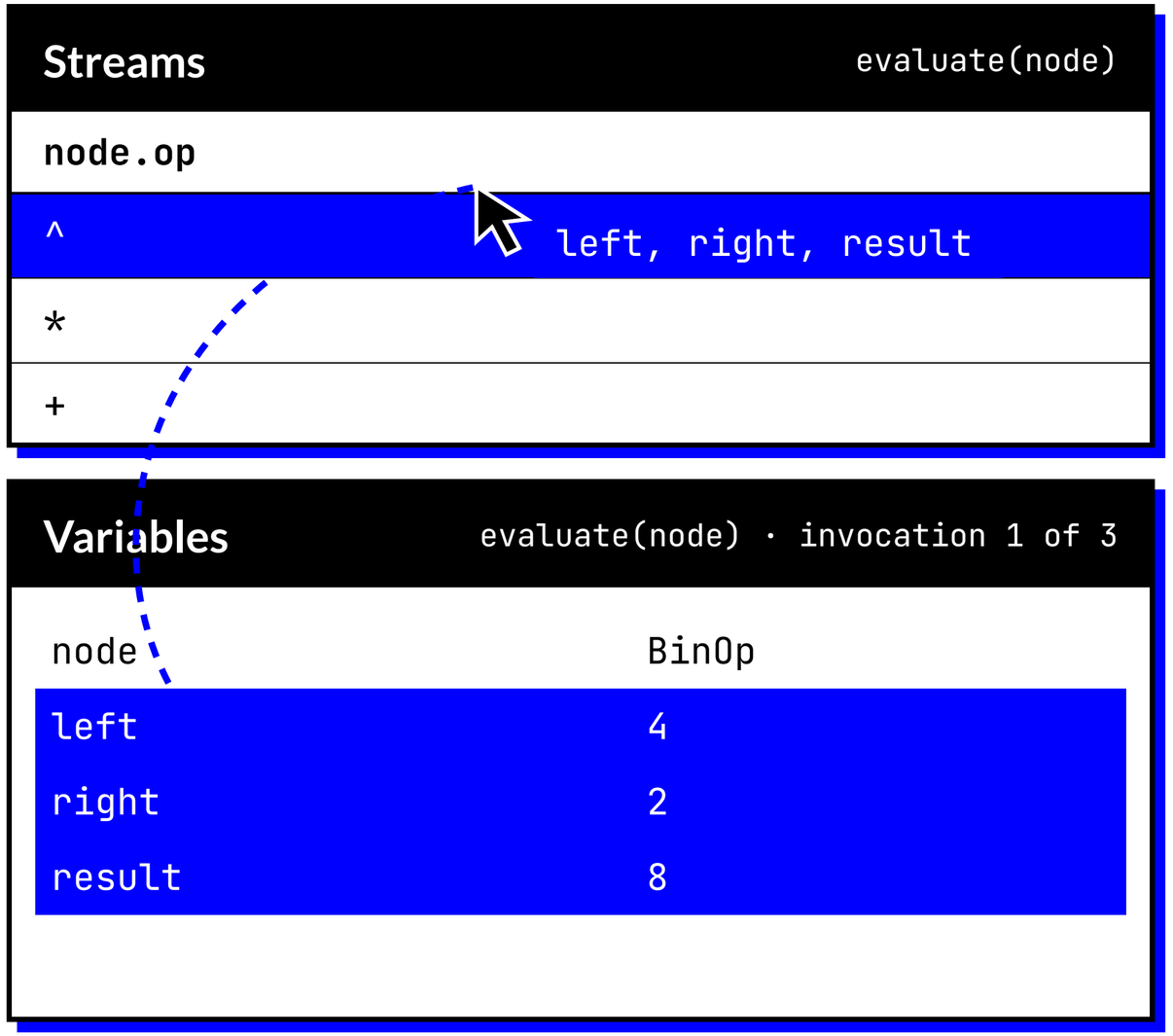

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     return result
```

Streams — evaluate(node)

node.op	left	right	result
^	4	2	8
*	3	8	24
+	2	24	26

Variables — evaluate(node) · invocation 1 of 3

node	BinOp
left	4
right	2
result	8

Data-Oriented Debugging Across Place and Time

Questions?

Jeffrey Tao

https://speculative.tech

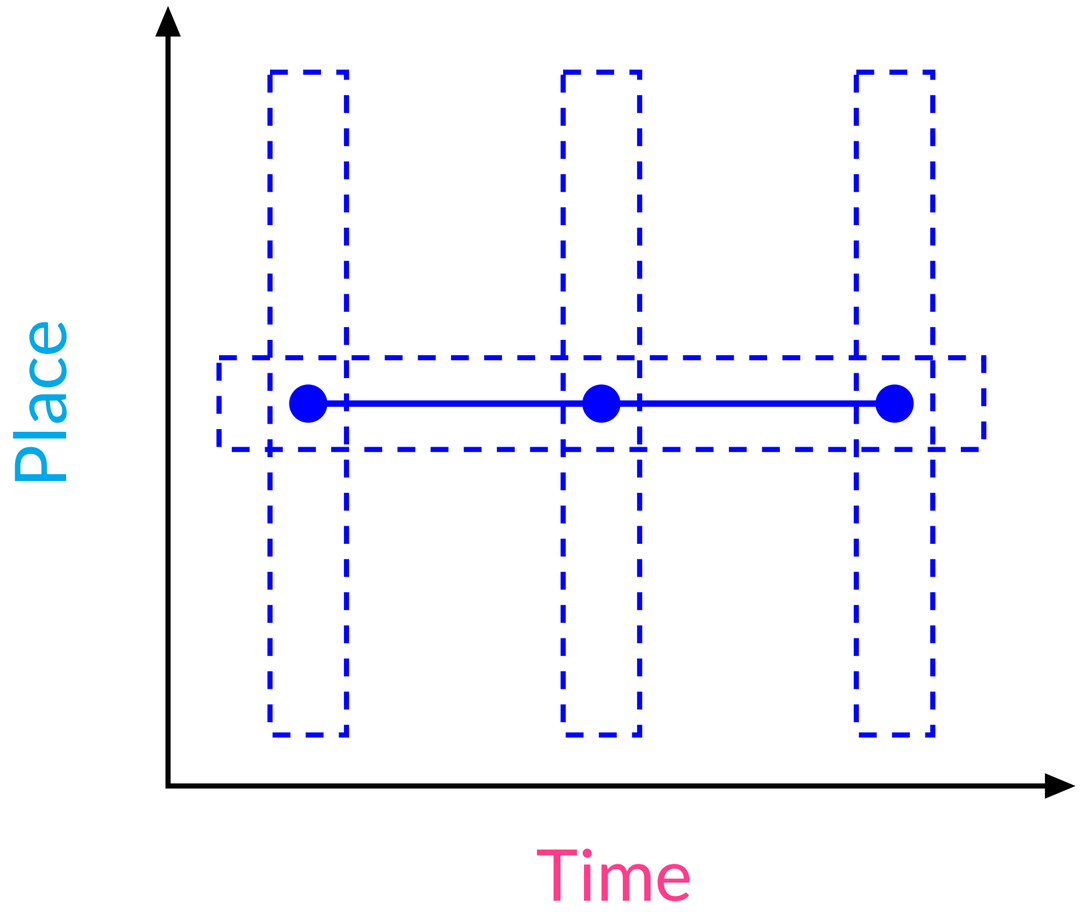
Appendix

🚨 Break glass in case of AI question

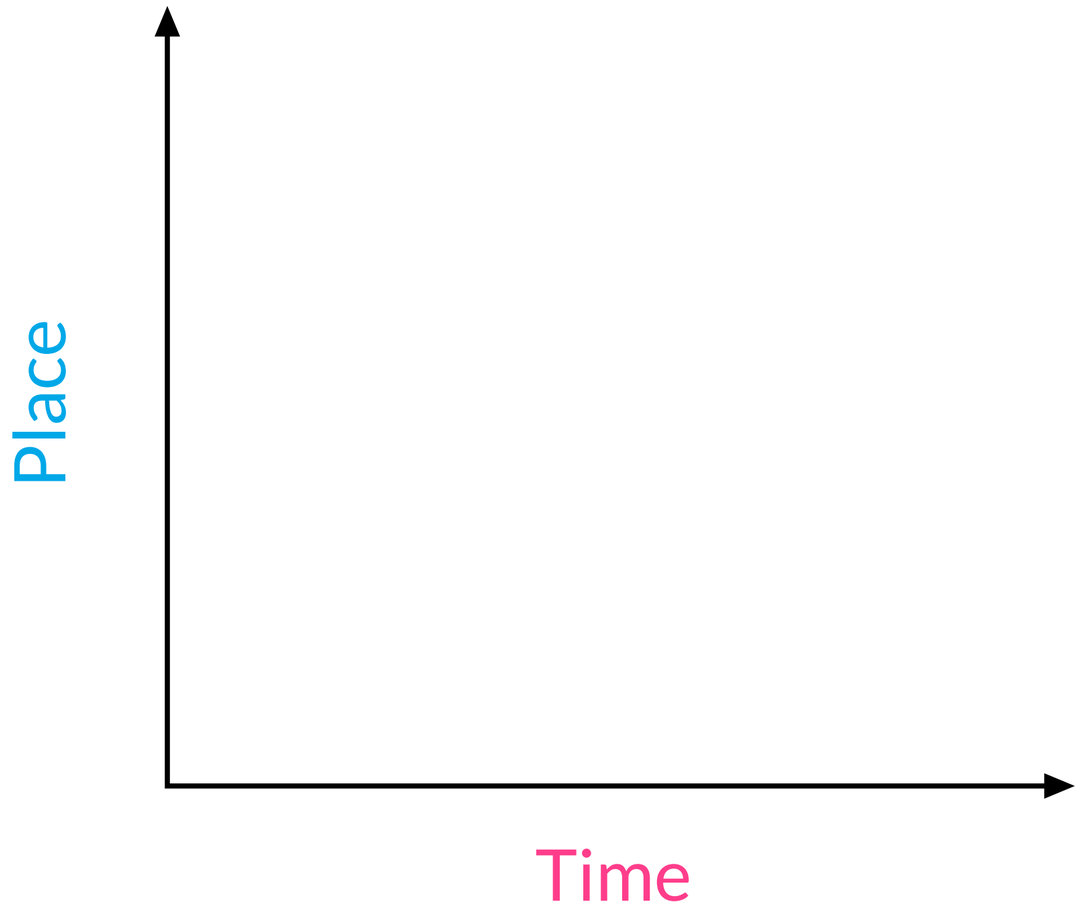
Hasn't AI made debugging interfaces obsolete?

- Data-Oriented Debugging concerns *responses to data needs*
- AI coding agents do better with objective feedback
- Debugging is still an interesting setting for observing human comprehension of complex systems

autopsy



autopsy

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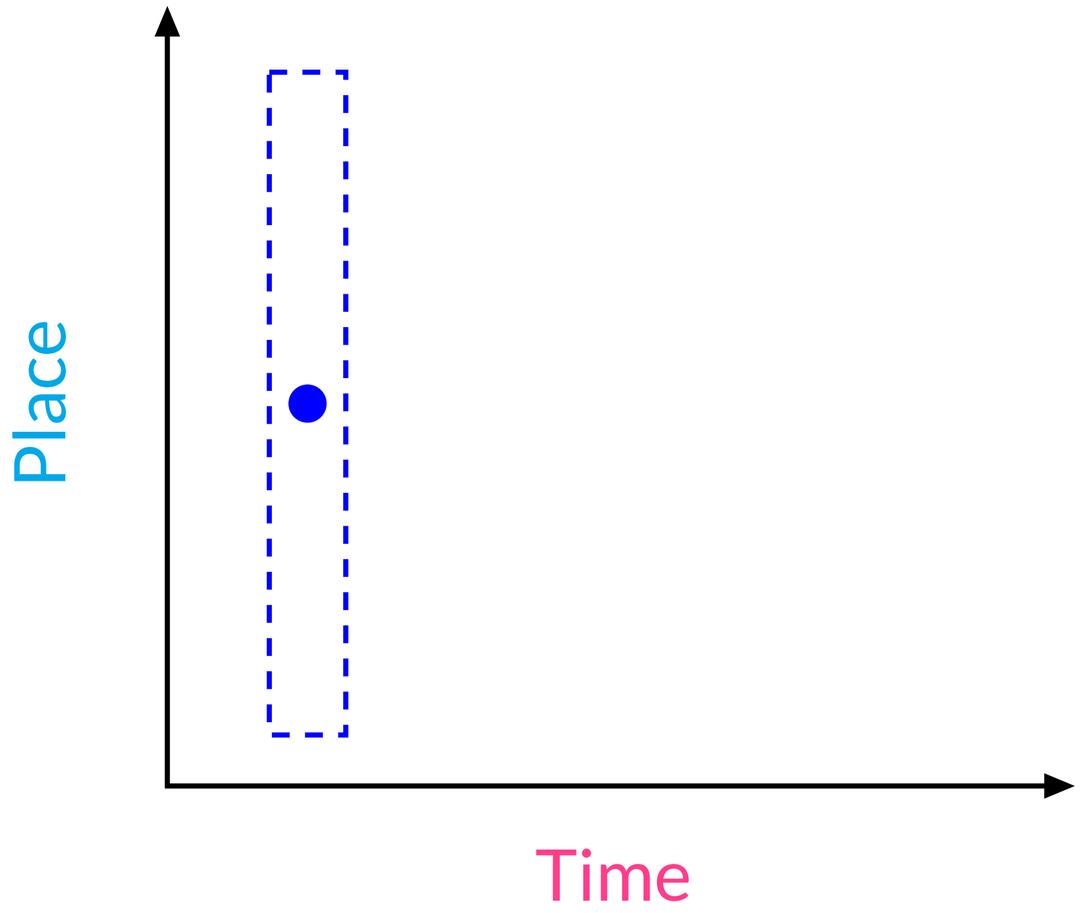
8         case "*": result = left * right
9         case "^": result = left * right
10     log(node.op)
11     return result
```

autopsy

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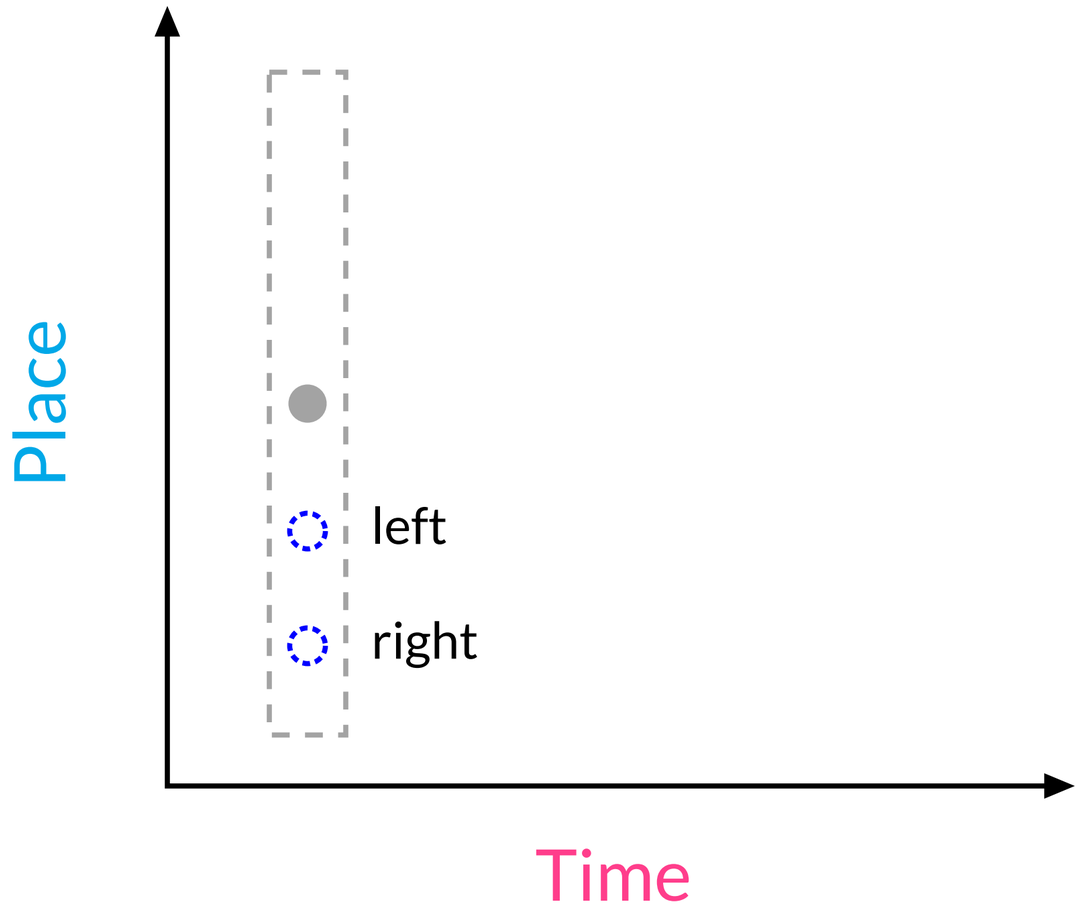
autopsy

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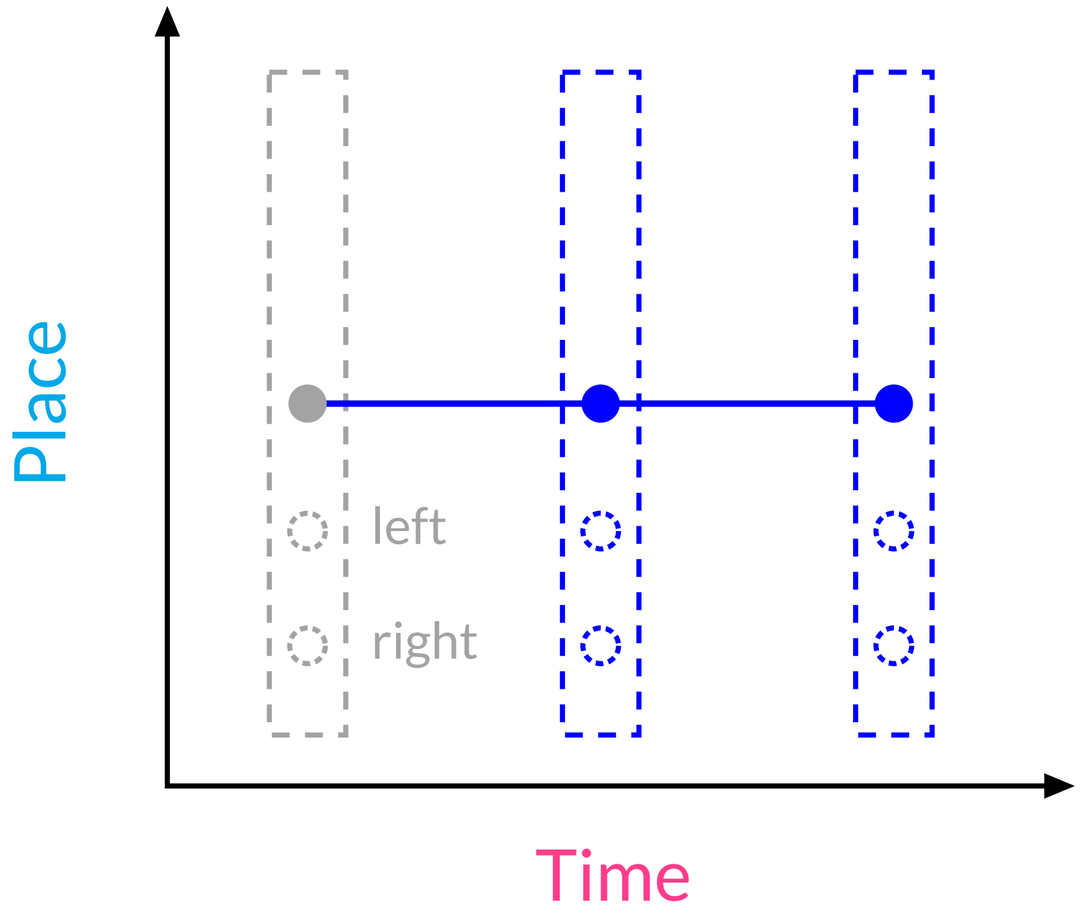
autopsy

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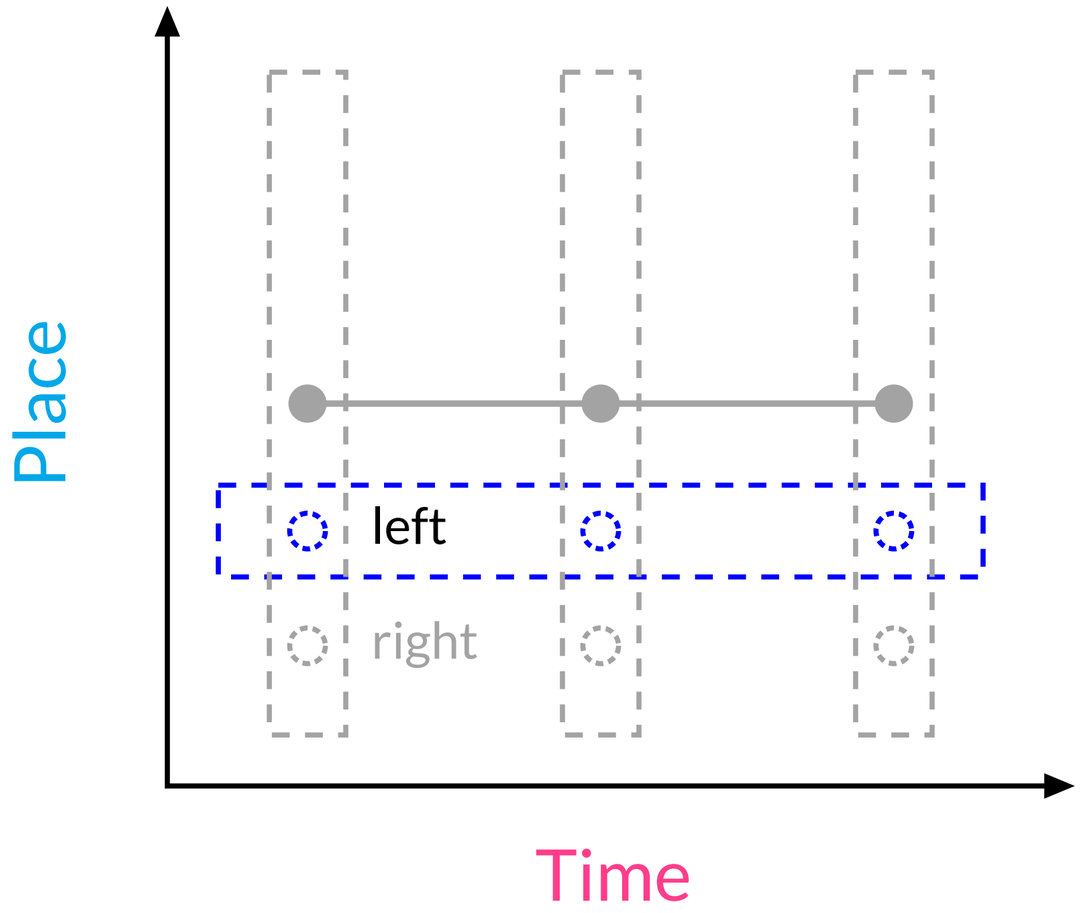
autopsy

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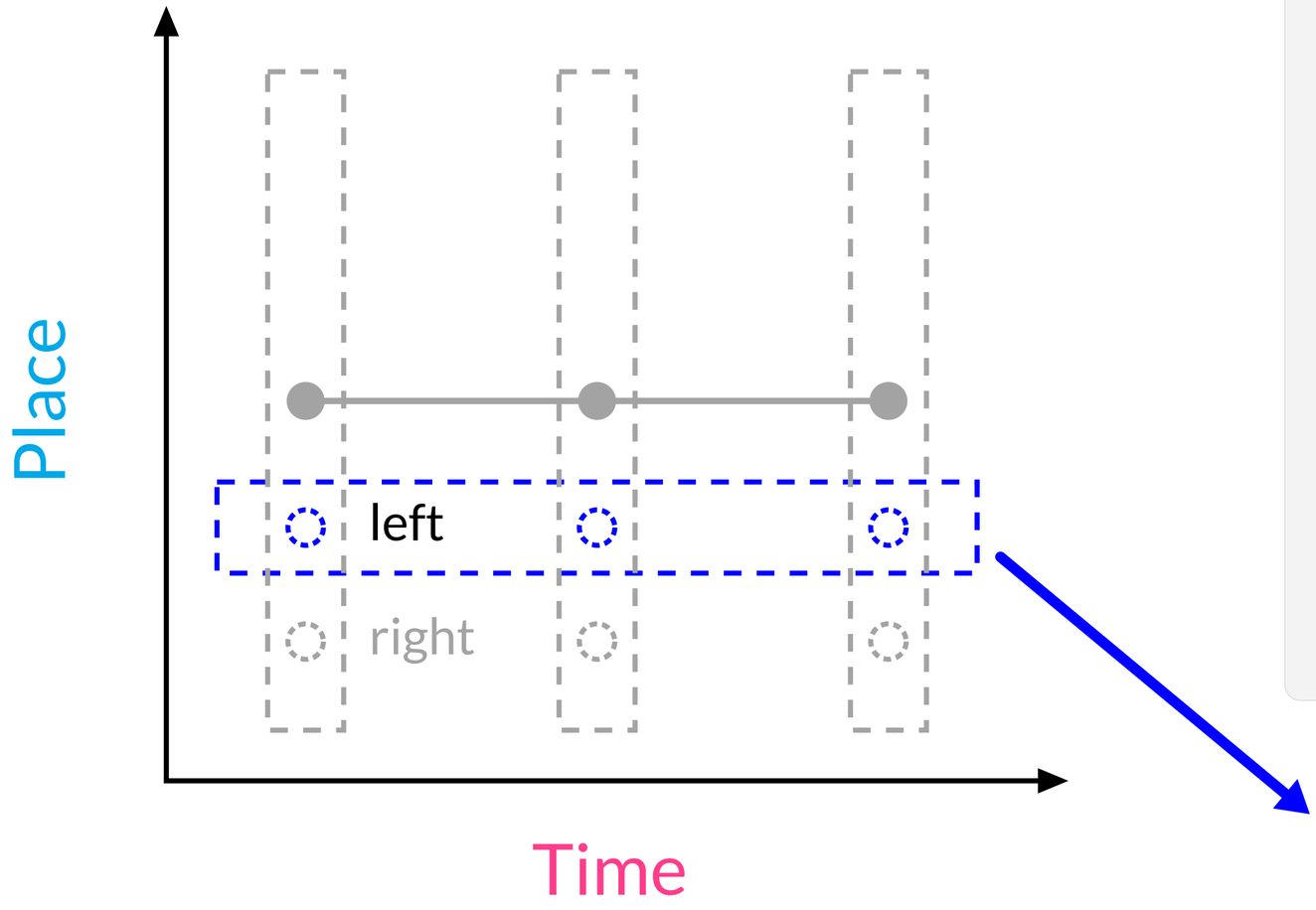


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min, avg, max, ...